Bubnov–Galerkin T-matrix for elastic scattering from a cubic inclusion

A systematic $T_9 \to T_{27} \to T_{57}$ hierarchy
with exact closed-form integrals

Research Notes

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1 Introduction

1.1 The physical problem

Consider a single cubic elastic inclusion $V = [-a, a]^3$ embedded in a homogeneous, isotropic background medium with P-wave speed α, S-wave speed β, and density ρ_0. The inclusion differs from the background by stiffness contrasts $\Delta\lambda$, $\Delta\mu$ and a density contrast $\Delta\rho$. An incident elastic plane wave (P, SV, or SH) impinges on the cube, generating a scattered field that encodes the inclusion's mechanical properties.

This configuration is fundamental to seismic scattering theory. Cubic heterogeneities arise naturally as:

- grain-scale models of crystalline aggregates,
- voxel representations of 3-D seismic tomography,
- idealised fracture zones bounded by planar surfaces.

In all cases the cube's discrete O_h symmetry (48 operations) is richer than a sphere's continuous $O(3)$ symmetry, but poor enough to break isotropy and generate shear-wave splitting.

1.2 The mathematical challenge

The standard approach to single-scatterer problems is the *volume integral equation* (VIE), also called the Lippmann–Schwinger (LS) equation. It expresses the total field inside the scatterer as a self-consistent integral against the free-space elastic Green's tensor G_{ij}. The difficulty lies in the singularity of G_{ij}:

$$G_{ij}^{\text{stat}}(\boldsymbol{s}) \sim \frac{1}{|\boldsymbol{s}|}, \quad \partial G \sim \frac{1}{|\boldsymbol{s}|^2}, \quad \partial\partial G \sim \frac{1}{|\boldsymbol{s}|^3}, \quad \partial\partial\partial G \sim \frac{1}{|\boldsymbol{s}|^4}. \tag{1}$$

The classical moment-method closure differentiates the LS equation at $\boldsymbol{r} = \boldsymbol{0}$ to obtain a finite system for the internal field. Already at first-derivative level, no integrable dominating function exists for the kernel $1/|\boldsymbol{s}|^2$ when $\boldsymbol{r}$ lies in the inclusion. At second-derivative level, the $1/|\boldsymbol{s}|^3$ singularity requires an Eshelby-type delta-function correction from the $r = 0$ principal-value decomposition. These distributional regularisations are algebraically involved and error-prone.

LS derivative order	worst kernel	status
0 (body force)	$1/\lvert\boldsymbol{s}\rvert$	L^1_{loc} ✓
0 (elastic)	$1/\lvert\boldsymbol{s}\rvert^2$	conditionally convergent (PV)
1 (body force)	$1/\lvert\boldsymbol{s}\rvert^2$	PV
1 (elastic)	$1/\lvert\boldsymbol{s}\rvert^3$	**Eshelby delta required**
2 (body force)	$1/\lvert\boldsymbol{s}\rvert^3$	**Eshelby delta required**
2 (elastic)	$1/\lvert\boldsymbol{s}\rvert^4$	**Hadamard finite part**

Table 1: Singular structure of the classical moment-method closure. Each derivative order introduces progressively non-integrable kernels.

1.3 The Bubnov–Galerkin solution

We replace point evaluation by a *Bubnov–Galerkin* (test = trial) weak projection onto a finite-dimensional polynomial space. The resulting bilinear forms involve only the Green's tensor G_{ij} itself—the singularity $1/|\boldsymbol{s}|$ is locally L^1 in $\mathbb{R}^3$—so *every entry of the closure matrix is an absolutely convergent integral.* No principal values, no delta corrections, no distributional bookkeeping.

The Galerkin projection, combined with the centre-of-mass/relative coordinate substitution on the cube, reduces all six-dimensional double integrals to three-dimensional *master integrals* that Mathematica evaluates in exact closed form.

1.4 Main result: the $T_9 \to T_{27} \to T_{57}$ hierarchy

The central result is a systematic hierarchy of frequency-dependent T-matrices with increasing accuracy, all computed *without numerical quadrature*:

Tier	Basis dim	Polynomial degree	Block sizes	Accuracy
T_9	9	≤ 1 (constant + linear)	max 1×1	Rayleigh limit
T_{27}	27	≤ 2 (+ quadratic)	max 4×4	$O((ka)^2)$ corrections
T_{57}	57	≤ 3 (+ cubic)	max 7×7	$O((ka)^4)$ corrections

Each tier adds the next polynomial degree to the trial space. The O_h symmetry of the cube forces the same seven irreducible representations at every tier; only the multiplicities (block sizes) grow. This makes the hierarchy both systematic and computationally tractable: the 57×57 T-matrix decomposes into seven independent blocks of size at most 7×7.

2 The Lippmann–Schwinger equation for elastic scattering

2.1 Setup

The background medium is homogeneous and isotropic with Lamé parameters λ_0, μ_0, density ρ_0, and wave speeds $\alpha = \sqrt{(\lambda_0 + 2\mu_0)/\rho_0}$ (P-wave), $\beta = \sqrt{\mu_0/\rho_0}$ (S-wave). The cubic inclusion $V = [-a, a]^3$ has stiffness contrast $\Delta c_{ijkl} = \Delta\lambda\, \delta_{ij}\delta_{kl} + \Delta\mu\, (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and density contrast $\Delta\rho$.

2.2 The integral equation

The Lippmann–Schwinger equation for the total displacement field is

$$u_i(\boldsymbol{r}) = u_i^0(\boldsymbol{r}) + \omega^2 \int_V G_{ij}(\boldsymbol{r} - \boldsymbol{r}')\, \Delta\rho\, u_j(\boldsymbol{r}')\, dV' - \int_V \partial'_k G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, \Delta c_{nklj}\, \partial'_l u_j(\boldsymbol{r}')\, dV', \qquad (2)$$

where G_{ij} is the free-space elastic Green's tensor and $\boldsymbol{u}^0$ the incident wave. Two source terms drive the scattering:

- a **body force** from the density contrast: $\omega^2 \Delta\rho\, \boldsymbol{u}$, and
- a **stress divergence** from the stiffness contrast: $\nabla \cdot (\Delta\mathbf{c} : \boldsymbol{\varepsilon}(\boldsymbol{u}))$.

2.3 Singularity structure

The static Green's tensor $G_{ij}^{\text{stat}}(\boldsymbol{s}) \sim 1/|\boldsymbol{s}|$ is locally integrable. Its first derivatives $\sim 1/|\boldsymbol{s}|^2$ are conditionally convergent (Cauchy principal value), and its second derivatives $\sim 1/|\boldsymbol{s}|^3$ require a delta-function correction at $\boldsymbol{s} = \mathbf{0}$ from the Gauss-law identity $\nabla^2(1/r) = -4\pi\delta(\boldsymbol{r})$. This is the *Eshelby singularity* that makes direct numerical quadrature of the stiffness channel unreliable.

The Galerkin formulation of Section 3 circumvents this problem entirely.

3 Bubnov–Galerkin discretisation and the $T_9 \to T_{27} \to T_{57}$ hierarchy

3.1 The Galerkin idea

We project eq. (2) onto a finite-dimensional trial space of vector polynomials on the cube. The test functions are chosen *equal to the trial functions* (Bubnov–Galerkin), producing symmetric bilinear forms. The key advantage is that integration by parts moves derivatives off the singular kernel G_{ij} onto the smooth polynomial trial functions, reducing the worst-case singularity from $1/|\boldsymbol{s}|^3$ to $1/|\boldsymbol{s}|$.

3.2 Trial space hierarchy

The trial functions have the form $\boldsymbol{\varphi}_j(\boldsymbol{r}) = c_m(\boldsymbol{r})\,\boldsymbol{\varepsilon}_k$, where c_m is a scalar polynomial and $\boldsymbol{\varepsilon}_k$ a Cartesian unit vector. The hierarchy is:

- T_9 **(degree ≤ 1):** 3 constant (displacement $\boldsymbol{\varepsilon}_k$) + 6 linear (strain $r_m\,\boldsymbol{\varepsilon}_k$, with symmetric gradient) = **9 basis functions**.
- T_{27} **(degree ≤ 2):** T_9 + 18 quadratic ($r_m^2\,\boldsymbol{\varepsilon}_k$ and $r_m r_n\,\boldsymbol{\varepsilon}_k$) = **27 basis functions**. The 3 rigid-rotation modes are removed by symmetrising the linear part.
- T_{57} **(degree ≤ 3):** T_{27} + 30 cubic ($r_m^3\,\boldsymbol{\varepsilon}_k$, $r_m^2 r_n\,\boldsymbol{\varepsilon}_k$, $r_m r_n r_p\,\boldsymbol{\varepsilon}_k$) = **57 basis functions**.

In general, each polynomial degree d adds $(d+1)(d+2)/2 \times 3$ new vector basis functions.

3.3 The Galerkin linear system

Definition 1 (Bubnov–Galerkin closure)**.** Find $\boldsymbol{u}^h \in T_N$ such that for all $\boldsymbol{\varphi} \in T_N$,

$$\langle \boldsymbol{\varphi}, \boldsymbol{u}^h \rangle_V = \langle \boldsymbol{\varphi}, \boldsymbol{u}^0 \rangle_V + \omega^2 \Delta\rho\, \mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}^h] - \Delta c_{nklj}\, \mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}^h], \tag{3}$$

where the bilinear forms are

$$\langle \boldsymbol{\varphi}, \boldsymbol{u} \rangle_V := \int_V \varphi_i(\boldsymbol{r})\, u_i(\boldsymbol{r})\, dV, \tag{4}$$

$$\mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, u_n(\boldsymbol{r}')\, dV\, dV', \tag{5}$$

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, \partial'_k G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, \partial'_l u_j(\boldsymbol{r}')\, dV\, dV'. \tag{6}$$

In matrix form, with coefficient vector $\boldsymbol{c}$ expanding $\boldsymbol{u}^h = \sum_\alpha c_\alpha\, \boldsymbol{\varphi}_\alpha$:

$$\big(M_N + B_N^{\text{LS}} - \varepsilon\, B_{\text{body},N}\big)\, \boldsymbol{c} = M_N\, \boldsymbol{c}^0, \tag{7}$$

where M_N is the mass (overlap) matrix, $B_{\text{body},N}$ the body bilinear, B_N^{LS} the LS-convolved stiffness, and $\varepsilon = \omega^2 \Delta\rho$.

3.4 Well-posedness

Proposition 2. *For polynomial $\boldsymbol{\varphi}, \boldsymbol{u} \in T_N$, every bilinear form in (3) is an absolutely convergent integral.*

Proof sketch. The mass form is trivial. For $\mathcal{B}_{\text{el}}$, integration by parts on $\boldsymbol{u}$ transfers derivatives from G to the polynomial:

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] = -\int_V\!\!\int_V \varphi_i\, G_{in}\, \partial'_k \partial'_l u_j\, dV'\, dV + \int_V \!\oint_{\partial V} \varphi_i\, G_{in}\, \partial'_l u_j\, n_k\, dS'\, dV. \tag{8}$$

Both integrands contain only $G_{in} \sim 1/|\boldsymbol{s}|$ (locally L^1 in $\mathbb{R}^3$) multiplied by polynomials. The six-fold integral $\iint_{V\times V} 1/|\boldsymbol{s}|\, dV\, dV'$ converges by the Hardy–Littlewood–Sobolev inequality. □ □

3.5 The T-matrix

The T-matrix relates incident-field coefficients to scattered-source coefficients:

$$\boldsymbol{c}^{\text{sc}} = T_N\, \boldsymbol{c}^0, \qquad T_N = \big(\varepsilon\, B_{\text{body}} - B^{\text{LS}}\big)\big(M + B^{\text{LS}} - \varepsilon\, B_{\text{body}}\big)^{-1}. \tag{9}$$

At T_9, this captures the Rayleigh limit exactly. Each subsequent tier adds higher-order corrections in ka, converging systematically as the polynomial degree increases.

4 The elastic Green's tensor: Kelvin + smooth decomposition

The full frequency-dependent Green's tensor decomposes as $\boldsymbol{G} = \boldsymbol{G}^K + \boldsymbol{G}^s$, separating the static singularity from the smooth dynamic correction.

4.1 Kelvin (static, singular) part

The static Green's tensor is

$$G^K_{ij}(\boldsymbol{s}) = a_0 \frac{\delta_{ij}}{|\boldsymbol{s}|} + b_0 \frac{s_i s_j}{|\boldsymbol{s}|^3}, \tag{10}$$

with Kelvin coefficients

$$a_0 = \frac{\alpha^2 + \beta^2}{8\pi\rho_0\alpha^2\beta^2}, \qquad b_0 = \frac{\alpha^2 - \beta^2}{8\pi\rho_0\alpha^2\beta^2}. \tag{11}$$

The a_0 term (proportional to δ_{ij}) is the **isotropic** A**-channel**; the b_0 term (proportional to $s_i s_j/|\boldsymbol{s}|^2$) is the **anisotropic** B**-channel**. For a Poisson solid ($\alpha/\beta = \sqrt{3}$), we have $b_0 = 2a_0$; in general $b_0/a_0 = (\alpha^2 - \beta^2)/(\alpha^2 + \beta^2)$.

4.2 Smooth (dynamic) part

The smooth remainder admits a Taylor expansion in even powers of the separation:

$$G^s_{ij}(\boldsymbol{s}) = \delta_{ij}\,\Phi(|\boldsymbol{s}|^2) + s_i s_j\,\Psi(|\boldsymbol{s}|^2), \qquad \Phi(u) = \sum_{n=0}^{\infty} \varphi_n\, u^n, \quad \Psi(u) = \sum_{n=0}^{\infty} \psi_n\, u^n, \tag{12}$$

where the Taylor coefficients φ_n, ψ_n are computed from the P- and S-wave spherical Bessel expansions. The leading terms are purely imaginary, providing *radiation damping* absent in the static-only approximation.

4.3 Why this decomposition

When the Kelvin part is substituted into the body bilinear form, the A-channel produces integrals against $1/|\boldsymbol{s}|$ and the B-channel against $s_i s_j/|\boldsymbol{s}|^3$—both reducing to the master integrals of Section 5. The smooth part replaces these singular kernels by $|\boldsymbol{s}|^{2n}$ and $s_i s_j\,|\boldsymbol{s}|^{2n}$, yielding *purely algebraic* (exact rational) integrals.

The total dynamic body bilinear is then

$$\mathcal{B}_{\text{body}}(\omega) = \underbrace{a_0\, B^{(A)} + b_0\, B^{(B)}}_{\text{static (Kelvin)}} + \sum_{n=0}^{N_T-1} \left[\varphi_n\, a^{2n}\, B^{(A)}_{\text{sm},n} + \psi_n\, a^{2n+2}\, B^{(B)}_{\text{sm},n}\right], \tag{13}$$

where $B^{(A,B)}_{\text{sm},n}$ are geometry-only matrices assembled from polynomial master integrals (exact rationals, no symbolic integration needed).

5 Analytical integration: master integrals on the cube

The central computational result is that *every* entry of the Galerkin closure reduces analytically to finite linear combinations of two families of three-dimensional master integrals over the unit cube $[0,1]^3$.

5.1 Centre-of-mass / relative coordinate substitution

The six-dimensional double integrals in (5) are reduced to three-fold integrals via the substitution

$$\boldsymbol{\xi} = \tfrac{1}{2}(\boldsymbol{r} + \boldsymbol{r}'), \qquad \boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}', \qquad d\boldsymbol{r}\, d\boldsymbol{r}' = d\boldsymbol{\xi}\, d\boldsymbol{s}. \tag{14}$$

For the cube $V = [-a, a]^3$, the constraint $\boldsymbol{r}, \boldsymbol{r}' \in V$ becomes $\boldsymbol{\xi} \in B(\boldsymbol{s})$, a centred box of half-lengths $b_i = a - |s_i|/2$. The inner $\boldsymbol{\xi}$-integral of any polynomial $P(\boldsymbol{\xi})$ is elementary (even powers survive, odd vanish by symmetry), yielding

$$\int_V\!\!\int_V P(\boldsymbol{r})\, K(\boldsymbol{r} - \boldsymbol{r}')\, Q(\boldsymbol{r}')\, d\boldsymbol{r}\, d\boldsymbol{r}' = \int_{[-2a,2a]^3} K(\boldsymbol{s}) \underbrace{\int_{B(\boldsymbol{s})} P(\boldsymbol{\xi} + \tfrac{\boldsymbol{s}}{2})\, Q(\boldsymbol{\xi} - \tfrac{\boldsymbol{s}}{2})\, d\boldsymbol{\xi}}_{\text{polynomial in } (\boldsymbol{s},a)}\, d\boldsymbol{s}. \tag{15}$$

For $P = Q = 1$, the inner integral is the *cube autocorrelation* (3-D tent function): $\Lambda_V(\boldsymbol{s}) = \prod_{i=1}^{3}(2a - |s_i|)_+$.

After rescaling $\boldsymbol{s} = 2a\boldsymbol{u}$ and exploiting octant symmetry $|u_i| \to u_i$, every entry reduces to integrals over $[0, 1]^3$ with a tent weight $(1 - u_1)(1 - u_2)(1 - u_3)$ times a monomial times one of two singular kernels.

5.2 A-channel master integrals

$$\mathcal{M}^{+}_{p,q,r} := \int_{[0,1]^3} \frac{x^p\, y^q\, z^r}{\sqrt{x^2 + y^2 + z^2}}\, dx\, dy\, dz \qquad (A\text{-channel}). \tag{16}$$

These are transcendental closed forms in algebraic numbers, π, logarithms, and inverse trigonometric functions. Cubic symmetry equates permutations: $\mathcal{M}^{+}_{p,q,r}$ depends only on the sorted triple $p \geq q \geq r$. The simplest examples:

$$\mathcal{M}^{+}_{0,0,0} = -\frac{\pi}{4} - \frac{1}{6}\log(70226 - 40545\sqrt{3}) \approx 1.1900387, \tag{17}$$

$$\mathcal{M}^{+}_{1,1,1} = \tfrac{1}{5}(1 - 4\sqrt{2} + 3\sqrt{3}) \approx 0.1078596. \tag{18}$$

5.3 B-channel master integrals

$$\mathcal{M}^{B+}_{p,q,r} := \int_{[0,1]^3} \frac{x^p\, y^q\, z^r}{(x^2 + y^2 + z^2)^{3/2}}\, dx\, dy\, dz \qquad (B\text{-channel}). \tag{19}$$

These carry the anisotropic $s_i s_j/|\boldsymbol{s}|^3$ kernel. The two families are linked by the trace identity

$$\mathcal{M}^{B+}_{p+2,q,r} + \mathcal{M}^{B+}_{p,q+2,r} + \mathcal{M}^{B+}_{p,q,r+2} = \mathcal{M}^{+}_{p,q,r}, \tag{20}$$

which follows from $|\boldsymbol{x}|^2/|\boldsymbol{x}|^3 = 1/|\boldsymbol{x}|$ and reduces the number of independent evaluations.

5.4 Polynomial (smooth) master integrals

When the smooth Green's tensor kernel replaces $1/|\boldsymbol{s}|$ by $|\boldsymbol{s}|^{2n}$, the master integral becomes purely algebraic:

$$\mathcal{M}^{\text{poly}}_{p,q,r}(n) := \int_{[0,1]^3} x^p y^q z^r\, (x^2{+}y^2{+}z^2)^n\, dx\, dy\, dz = \sum_{\substack{a+b+c=n \\ a,b,c\geq 0}} \frac{n!}{a!\, b!\, c!}\, \frac{1}{(p{+}2a{+}1)(q{+}2b{+}1)(r{+}2c{+}1)}, \tag{21}$$

an exact rational number for any non-negative integers p, q, r, n.

5.5 Tent-convolved integrals

The tent weight $(1-u_1)(1-u_2)(1-u_3)$ from the centre-of-mass reduction is absorbed by expanding as an eight-term alternating sum:

$$K_{1,\mathrm{at}}[p,q,r] = \sum_{\epsilon\in\{0,1\}^3} (-1)^{|\epsilon|}\, \mathcal{M}^{+}_{p+\epsilon_1,\, q+\epsilon_2,\, r+\epsilon_3}, \tag{22}$$

and similarly for the B-channel, reducing each body bilinear entry to a signed combination of $\mathcal{M}^{+}$ (or $\mathcal{M}^{B+}$) values.

5.6 Surface stiffness integrals

The LS-convolved stiffness (6), after integration by parts, splits into a volume force (reducible to existing body integrals) and a surface traction integral. For each face of the cube, the (σ,ξ,v) substitution

$$\sigma_a = \tfrac{1}{2}(r_a+s_a), \quad \xi_a = \tfrac{1}{2}(r_a-s_a), \quad v = \tfrac{1}{2}(1-r_k), \tag{23}$$

reduces the five-dimensional integral to three dimensions over $(\xi_1,\xi_2,v)\in[0,1]^3$, where each entry becomes a linear combination of $\mathcal{M}^{+}_{p,q,r}$ and $\mathcal{M}^{B+}_{p,q,r}$ values:

$$B^{\mathrm{LS,surf}}_{\alpha\beta}\big|_{\mathrm{face}} = 64\sum_{p,q,r} c^{(A)}_{pqr}\, \mathcal{M}^{+}_{p,q,r} + 16\sum_{p,q,r} c^{(B)}_{pqr}\, \mathcal{M}^{B+}_{p,q,r}. \tag{24}$$

No numerical quadrature is required at any stage.

5.7 Tables of master integrals

5.8 Verified analytical master integrals

The verification at degree 0 was reported in Section 5.2. We extend the table to include all parity-allowed master integrals up to degree 6:

(p,q,r)	$\mathcal{M}_{p,q,r} = \int_{[-1,1]^3} \frac{x^p y^q z^r}{\lvert\boldsymbol{r}\rvert}\, dV$	numerical
$(0,0,0)$	$-2\pi - \frac{4}{3}\log(70226-40545\sqrt{3})$	9.5203095
$(2,0,0)$	$\frac{1}{9}\big(6\sqrt{3}-\pi+\log(3650401+2107560\sqrt{3})\big)$	2.5615786
$(2,2,0)$	$\frac{1}{135}\big(108\sqrt{3}+18\pi+63\log 2+20\log(26-15\sqrt{3})$	0.75095335
	$-60\log(\sqrt{3}-1)-66\log(1+\sqrt{3})-90\coth^{-1}\sqrt{3}\big)$	
$(4,0,0)$	$-\frac{2}{45}\big(12\sqrt{3}+7\pi-42\log 2$	1.4351485
	$+99\log(\sqrt{3}-1)-15\log(1+\sqrt{3})\big)$	
$(2,2,2)$	$\frac{1}{1260}\big(1008\sqrt{3}-42\pi+522\log 2+441\log(\sqrt{3}-1)$	0.22735457
	$-1395\log(1+\sqrt{3})-84\log(2+\sqrt{3})-90\log(11\sqrt{3}-19)$	
	$+35\log(26+15\sqrt{3})-672\coth^{-1}\sqrt{3}\big)$	
$(4,2,0)$	$\frac{1}{630}\big(-48\sqrt{3}+17\pi$	0.42942069
	$+12\tanh^{-1}\frac{21252634831\sqrt{3}}{36810643322}\big)$	
$(6,0,0)$	$\frac{1}{42}\big(-5\pi-4\log 2+8(3\sqrt{3}+\log(5+3\sqrt{3}))\big)$	0.99201788

Table 2: Closed-form values of the scalar master integrals $\mathcal{M}_{p,q,r}$ over the unit cube. Cubic symmetry equates entries that are permutations of each other; only canonical orderings $p\geq q\geq r$ are listed.

The same family of integrals over the positive octant $[0,1]^3$ supplies the body-bilinear-form building blocks; we collect them in Table 3.

(p,q,r)	$\mathcal{M}^{+}_{p,q,r} = \int_{[0,1]^3} \frac{x^p y^q z^r}{\lvert \boldsymbol{r} \rvert}\, dV$	numerical
$(0,0,0)$	$-\frac{\pi}{4} - \frac{1}{6}\log\big(70226 - 40545\sqrt{3}\big)$	1.1900387
$(1,0,0)$	$\frac{1}{36}\big(-2\pi - 12\big(\sqrt{2} - \sqrt{3} + \sinh^{-1}(1)\big)$ $+ \cosh^{-1}(189750626) + 18\coth^{-1}\sqrt{3}\big)$	0.51559356
$(1,1,0)$	$\frac{1}{12}\big(1 - 7\sqrt{2} + 6\sqrt{3} + \log(5\sqrt{2} - 7)$ $- 3\log(2 - \sqrt{3})\big)$	0.23329690
$(2,0,0)$	$\frac{1}{144}\big(12\sqrt{3} - 2\pi$ $+ \log(26650854921601 + 15386878263120\sqrt{3})\big)$	0.32019732
$(1,1,1)$	$\frac{1}{5}\big(1 - 4\sqrt{2} + 3\sqrt{3}\big)$	0.10785963
$(2,1,0)$	$\frac{1}{2880}\big(-504\sqrt{2} + 672\sqrt{3} + 32\pi + 72\log(\sqrt{3} - 1)$ $- 252\log(1 + \sqrt{3}) + 90\log(2(2 + \sqrt{3})) - 261\sinh^{-1}(1)$ $+ 40\cosh^{-1}(26) + 45\coth^{-1}\sqrt{2}\big)$	0.14741068

Table 3: Closed-form values of the positive-octant master integrals $\mathcal{M}^{+}_{p,q,r}$ over $[0,1]^3$. Up to canonical ordering $p \geq q \geq r$ and $p + q + r \leq 3$ they exhaust all polynomial weights needed by the Path-B body bilinear form on the constant-and-linear subspace.

5.9 Body bilinear-form building blocks

For the constant subspace of T_{27} the diagonal and off-diagonal entries of the body bilinear form $\mathcal{B}_{\text{body}}$ reduce, after the centre-of-mass / relative-coordinate substitution and an octant-symmetry restriction, to three positive scalars. Define

$$i_A := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{1}{|\boldsymbol{u}|}\, d\boldsymbol{u}, \tag{25}$$

$$i_{B,11} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}, \tag{26}$$

$$i_{B,12} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1 u_2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}. \tag{27}$$

The first reduces by linearity to a signed combination of $\mathcal{M}^{+}_{p,q,r}$ values (Table 3):

$$i_A \;=\; \mathcal{M}^{+}_{0,0,0} \;-\; 3\mathcal{M}^{+}_{1,0,0} \;+\; 3\mathcal{M}^{+}_{1,1,0} \;-\; \mathcal{M}^{+}_{1,1,1} \;\approx\; 0.2352890805487075. \tag{28}$$

The remaining two are obtained by integrating once by parts in the u_1 direction; the singular kernel $|\boldsymbol{u}|^{-3}$ collapses to a $|\boldsymbol{u}|^{-1}$-class kernel and a clean two-dimensional residual:

$$i_{B,11} \;=\; i_A \;-\; J_{11}, \qquad J_{11} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\Big[\sqrt{1+u_2^2+u_3^2} - \sqrt{u_2^2+u_3^2}\Big]\, du_2\, du_3, \tag{29}$$

$$i_{B,12} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\, u_2 \Big[\frac{1}{\sqrt{u_2^2+u_3^2}} \;-\; \operatorname{arcsinh}\Big(\frac{1}{\sqrt{u_2^2+u_3^2}}\Big)\Big]\, du_2\, du_3. \tag{30}$$

Mathematica evaluates both two-dimensional integrals in closed form:

$$J_{11} \;=\; \frac{1}{240}\Big(8 + 8\sqrt{2} - 16\sqrt{3} - 55\log(2) + 15\log\big(3+2\sqrt{2}\big) + 110\log\big(1+\sqrt{3}\big) - 15\log\big(2+\sqrt{3}\big) - 80\tan^{-1}\Big(\frac{3}{8}\Big) + 80\tan^{-1}\Big(\frac{1}{183}\big(96 - 73\sqrt{3}\big)\Big) + 10\sinh^{-1}(1)\Big)$$

with $J_{11} \approx 0.1568593870324717$, and

$$i_{B,12} \;=\; \frac{1}{240}\Big(-8\sqrt{2} + 4\sqrt{3} + 20\log(2) + 5\log(256) + 40\log\big(1+\sqrt{2}\big) + 22\log\big(3+2\sqrt{2}\big) - 120\log\big(1+\sqrt{3}\big) - 22\log\big(2+\sqrt{3}\big) + 20\tan^{-1}\Big(\frac{1}{2}\Big) + 16\tan^{-1}\Big(\frac{5}{8}\Big) - 40\tan^{-1}\Big(\frac{3}{2}\Big) + 40\tan^{-1}\Big(8 + \frac{13}{\sqrt{3}}\Big) - 20\tan^{-1}\Big(\frac{1}{11}\big(8 - 5\sqrt{3}\big)\Big) - 16\tan^{-1}\Big(\frac{1}{167}\big(160 + 89\sqrt{3}\big)\Big) + 4\sinh^{-1}(1) + 36\sinh^{-1}\Big(\frac{1}{\sqrt{2}}\Big)\Big)$$

with $i_{B,12} \approx 0.04970171296728090$.

A non-trivial consistency check is supplied by the trace identity

$$3\, i_{B,11} \;=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2+u_2^2+u_3^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u} \;=\; i_A,$$

which is verified analytically and numerically to working precision.

5.10 The 9×9 body bilinear matrix on T_9

With the master integrals of Sections 5.8 and the three scalar building blocks i_A, $i_{B,11}$, $i_{B,12}$, the body bilinear form $\mathcal{B}_9 : T_9 \times T_9 \to \mathbb{R}$ is assembled entry-by-entry from

$$\mathcal{B}_9[i,j] \;=\; 32 \sum_{p,q=1}^{3} \Big\{ A\, \delta_{pq}\, K_1[\Pi_{ij,pq}] \;+\; B\, K_3[\Pi_{ij,pq};\, p,q] \Big\}, \tag{31}$$

where $\Pi_{ij,pq}(\boldsymbol{u})$ is the residual polynomial obtained by (a) substituting $\phi_p^{(i)}(\boldsymbol{r})\phi_q^{(j)}(\boldsymbol{r}')$, (b) integrating the centre-of-mass variable $\boldsymbol{\xi}$ over the overlap box, and (c) dividing out the standard tent factor $(1-u_1)(1-u_2)(1-u_3)$. The prefactor $32 = 4 \cdot 8/1$ absorbs the $s \mapsto 2u$ Jacobian 8, the Green-tensor rescaling $G(2\boldsymbol{u}) = \frac{1}{2}[A/|\boldsymbol{u}| + B\, u_p u_q/|\boldsymbol{u}|^3]$, and the symmetric-box factor 2/1. A further factor of 8 appears on parity-even entries when we restrict to the positive octant $[0,1]^3$, so a diagonal displacement-block entry is proportional to 256.

Under cubic symmetry the T_9-basis decomposes into four irreducible blocks, each of which is characterised by at most two independent scalars:

- **Displacement block** ($i,j \in \{1,2,3\}$): diagonal, with a single scalar.
- **Displacement–strain block** ($i \in \{1,2,3\}$, $j \in \{4,\dots,9\}$): identically zero by parity.
- **Axial-strain block** ($i,j \in \{4,5,6\}$): diagonal entry α, off-diagonal entry β.
- **Axial–shear block** ($i \in \{4,5,6\}$, $j \in \{7,8,9\}$): identically zero by parity.
- **Shear-strain block** ($i,j \in \{7,8,9\}$): diagonal entry γ, off-diagonals identically zero by parity.

Displacement block

All three displacement-block diagonal entries are equal and reduce in closed form to

$$\mathcal{B}_9[1,1] \;=\; \mathcal{B}_9[2,2] \;=\; \mathcal{B}_9[3,3] \;=\; 256\,\big(A\, i_A \;+\; B\, i_{B,11}\big) \;=\; 60.23400462046913\, A \;+\; 20.07800154015638\, B, \tag{32}$$

with off-diagonal entries vanishing by parity.

Axial-strain block

The diagonal entry $\alpha := \mathcal{B}_9[4,4] = \mathcal{B}_9[5,5] = \mathcal{B}_9[6,6]$ splits into

$$\alpha \;=\; \tfrac{256}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) \;+\; B\big(i_{B,11} - 2K_3^{(2,0,0)}\big)\Big] \;=\; 15.37252824403423\, A \;+\; 3.949761457945144\, B, \tag{33}$$

where

$$K_1^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2}{|\boldsymbol{u}|}\, d\boldsymbol{u}, \quad K_3^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^4}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}.$$

The off-diagonal entry $\beta := \mathcal{B}_9[4,5]$ is purely a B-channel correction,

$$\beta \;=\; -256\, B\, K_{3,\text{off}}^{(1,1,0)} \;=\; 0\, A \;-\; 1.471925680511371\, B, \tag{34}$$

where $K_{3,\text{off}}^{(1,1,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, u_1 u_2\, \frac{u_1 u_2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}$. Note that β vanishes in the A-channel because the integrand of the A piece is odd in (u_1, u_2).

Shear-strain block

The diagonal entry $\gamma := \mathcal{B}_9[7,7] = \mathcal{B}_9[8,8] = \mathcal{B}_9[9,9]$ evaluates to

$$\gamma \;=\; \tfrac{128}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) + B\big(i_{B,11} - 2K_3^{(0,2,0)}\big)\Big] \;-\; 128\, B\, K_{3,\text{off}}^{(1,1,0)} \;=\; 7.686264122017116\, A \;+\; 2.119728856266587\, B, \tag{35}$$

where $K_3^{(0,2,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_2^2 u_1^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}$. The relation $\gamma^{(A)} = \frac{1}{2}\,\alpha^{(A)}$ is an exact identity: the A-channel of γ is exactly half the A-channel of α, coming from the bilinearity of the two-component shear basis $\phi^{(7)} = (0, r_3/2, r_2/2)$.

Numerical verification of cubic symmetry (all six independent identities) was performed to 10^{-20} precision; see the Mathematica script output.

5.11 The 18×18 quad–quad block on T_{27}

Extending T_9 to T_{27} adjoins the 18 quadratic vector fields

$$\phi^{(9+6(k-1)+m)}(\boldsymbol{r}) \;=\; q_m(\boldsymbol{r})\, \boldsymbol{\varepsilon}_k, \qquad k \in \{1,2,3\},\;\; m \in \{1,\ldots,6\},$$

where the six quadratic monomials in canonical Voigt order are

$$q_1 = r_1^2, \quad q_2 = r_2^2, \quad q_3 = r_3^2, \quad q_4 = r_2 r_3, \quad q_5 = r_1 r_3, \quad q_6 = r_1 r_2.$$

We refer to $q_{1..3}$ as the squared (S) modes and to $q_{4..6}$ as the cross (X) modes. On the 18×18 quad–quad sub-block, the body bilinear form decomposes into an A-channel and a B-channel,

$$\mathcal{B}_{27}^{\text{quad}}[i,j] \;=\; 32\,\big(a_{ij}\, A \;+\; b_{ij}\, B\big), \tag{36}$$

where the scalars a_{ij}, b_{ij} depend only on the orbit of (i,j) under the combined O_h action on the three coordinate axes and the $i \leftrightarrow j$ symmetry.

Enumerating all $18^2 = 324$ index pairs and partitioning by this action yields exactly **34 orbits**. Of these, **20 orbits** (total multiplicity 222) are identically zero in both channels: each such vanishing is a parity cancellation forced by cubic symmetry (the integrand is odd in at least one coordinate axis under the $[-1,1]^3$ integration). The remaining **14 orbits** (total multiplicity 102) carry nontrivial values, and are listed in Table 4.

Independent scalars. The 13 non-zero orbits collapse to **3 distinct A-scalars and 9 distinct B-scalars**:

$$\begin{array}{lll}
\alpha_1 \approx 17.126396135464, & \alpha_2 \approx 14.698679523965, & \alpha_3 \approx 3.988696969908, \\
\beta_1 \approx 7.086377754857, & \beta_2 \approx 5.020009190303, & \beta_3 \approx 5.444954908314, \\
\beta_4 \approx 3.808769707336, & \beta_5 \approx 1.644075434840, & \beta_6 \approx 1.172310767534, \\
\beta_8 \approx 0.654189191338, & \beta_9 \approx -0.389578544263, & \beta_{10} \approx -0.216414577671.
\end{array}$$

The numbering omits β_7 deliberately: the orbit $(11,17)$ that previously appeared to give $\beta_7 \approx 0.692769$ is, at `WorkingPrecision`$\to 30$, identically zero in both channels (see paragraph below and Section 5.12). Across all 324 entries the observed within-orbit spread is 0 exactly in the A-channel and $< 10^{-27}$ in the B-channel, so these 12 scalars completely determine the 18×18 block. All values are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl`.

Representative (i,j)	Structure	mult	A-value	B-value
$(10,10)$	S-S diag, axis-matched	3	17.126396135464	7.086377754857
$(11,11)$	S-S diag, axis-unmatched	6	17.126396135464	5.020009190303
$(10,11)$	S-S same-block off-diag, one matched	12	14.698679523965	5.444954908314
$(11,12)$	S-S same-block off-diag, both unmatched	6	14.698679523965	3.808769707336
$(13,13)$	X-X diag, no-factor-matched	3	3.988696969908	1.644075434840
$(14,14)$	X-X diag, one-factor-matched	6	3.988696969908	1.172310767534
$(10,17)$	S-S cross-block (type Ia)	6	0	0.654189191338
$(11,16)$	S-S cross-block (type Ib)	6	0	0.654189191338
$(14,26)$	X-X cross-block (type I)	6	0	0.654189191338
$(13,20)$	X-X cross-block (type II)	6	0	−0.389578544263
$(14,19)$	X-X cross-block (type III)	6	0	−0.389578544263
$(10,21)$	S-X cross-block (type II)	12	0	−0.216414577671
$(11,21)$	S-X cross-block (type III)	12	0	−0.216414577671

Table 4: The 13 non-vanishing orbits of the 18×18 quad–quad block $\mathcal{B}_{27}^{\text{quad}}$. Indices i,j are global ($10 \le i,j \le 27$). The A- and B-values in each row are the two numerical coefficients of A and B in (36). Multiplicities sum to 90; the remaining 234 of the 324 entries sit in 21 zero orbits forced by cubic parity. Values are exact to all 12 digits displayed; the underlying high-precision data (`WorkingPrecision`→ 30, see Section 5.12) carries ∼ 28 correct digits.

Symmetry structure and accidental identities. The A-channel vanishes on every cross-direction orbit ($k_i \neq k_j$) and on every S-X orbit: this reflects the fact that the A part of the elastic Green's tensor is proportional to $\delta_{ij}/|\mathbf{r}-\mathbf{r}'|$ (an isotropic trace piece), so it only connects modes carrying identical Cartesian vector components and with an even polynomial overlap. The B-channel, which carries the $(\mathbf{r}-\mathbf{r}')_i(\mathbf{r}-\mathbf{r}')_j/|\mathbf{r}-\mathbf{r}'|^3$ anisotropic piece, is nonzero on all 14 orbits.

An additional structural observation from Table 4 is that the 7 surviving cross-direction orbits collapse to only *three* distinct B-values: $\beta_8 = 0.6542$ (appearing in three orbits), $\beta_9 = -0.3896$ (appearing in two orbits), and $\beta_{10} = -0.2164$ (appearing in two orbits). These collapses are identities to $\geq$ 27-digit precision (`WorkingPrecision`→ 30) that are not a direct consequence of the O_h-orbit decomposition alone: our canonical labelling distinguishes $(10,17)$, $(11,16)$, and $(14,26)$ as separate O_h orbits, yet they integrate to the same number. Together with the unconditional vanishing of orbit $(11,17)$ (described next), they reflect a deeper structural identity of the cube Green-tensor double integral; closed-form verification via PSLQ on the high-precision values of Section 5.12 is the natural next step.

The vanishing of orbit $(11,17)$**.** A separate, qualitative finding of the high-precision recomputation is that the S-X cross-block orbit at $(11,17)$ (S-X type I, multiplicity 12) is *exactly zero in both channels*. Earlier `MachinePrecision` assemblies returned $B \approx 0.692769$ for this orbit, which we previously labelled β_7; the overnight `WorkingPrecision`→ 30 recomputation collapses this value to $|B| < 10^{-27}$, consistent with an exact algebraic cancellation rather than residual noise. We have therefore reclassified $(11,17)$ as a zero orbit and dropped β_7 from the list of independent scalars. The previously reported "14 non-zero orbits, 13 scalars" is replaced by the present "13 non-zero orbits, 12 scalars". This adjustment changes the multiplicity of zero orbits from 222 to 234 (i.e. $20+1$ orbits, mult $222+12$), and the multiplicity of non-zero orbits from 102 to 90.

Extraction method and precision. The scalars α_i, β_j were initially computed by `Mathematica/CubeT6Block.wl`, which assembles $\mathcal{B}_{27}^{\text{quad}}$ via `NIntegrate` at `MachinePrecision` with `PrecisionGoal`→ 12. That configuration completes all 324 entries in ∼ 150 s and cross-checks against the T_9 Tier-5 displacement diagonal (32) to better than 3×10^{-12}, but it is just

at the edge of double precision: the trailing 1–3 decimal digits of the α_i, β_j are noise, and orbit $(11, 17)$ in particular is misreported as having $B \approx 0.6928$ rather than its true value 0.

The dedicated overnight high-precision recomputation `Mathematica/CubeT6OvernightHighPrec.wl` ran the same 14 orbit representatives at `WorkingPrecision` $\to 30$ on 2026-04-08, completing in ~ 1217 s of total `NIntegrate` time. Its results are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl` and tabulated in Section 5.12. The high-precision pass:

(P1) confirms the 12-digit values of the surviving 13 orbits (corrections at the 11th–12th decimal where machine-precision drift was visible);

(P2) shrinks the within-orbit spread of the α- and β-channels to $< 10^{-27}$, raising the cubic-symmetry consistency check from $\sim 10^{-12}$ to $\sim 10^{-27}$;

(P3) promotes orbit $(11, 17)$ from "$\beta_7 \approx 0.6928$" to a confirmed structural zero;

(P4) supplies the ≥ 28 correct digits needed for PSLQ-based closed-form reconstruction of the α_i and β_j in the same spirit as (33)–(35).

The 7 exact high-degree masters $\mathcal{M}^+_{(2,2,2)}, \ldots, \mathcal{M}^+_{(8,0,0)}$ primed from `CubeT6Masters.wl` continue to anchor the leading parity-even contributions in closed form; PSLQ reconstruction of the remaining α_i, β_j in terms of $\sqrt{3}$, π, log, $\tan^{-1}$, and $\sinh^{-1}$ atoms is the next analytical stage.

5.12 High-precision scalar values

The values below are produced by `Mathematica/CubeT6OvernightHighPrec.wl` at `WorkingPrecision`$\to$ 30 and stored verbatim in `Mathematica/CubeT6ScalarValues_HighPrec.wl`. Each entry carries ≥ 28 correct decimal digits and is suitable as input to PSLQ-style integer-relation searches against an atom basis $\{1, \sqrt{3}, \pi, \log 2, \log(1+\sqrt{3}), \log(\sqrt{3}-1), \sinh^{-1} 1, \tan^{-1}(\cdot), \ldots\}$ in the spirit of Tables 2–3.

Symbol	Value (28 digits)	Source orbit
α_1	17.1263961354644384869032245660	$(10, 10), (11, 11)$
α_2	14.6986795239648355731322989855	$(10, 11), (11, 12)$
α_3	3.9886969699081621215723702580	$(13, 13), (14, 14)$
β_1	7.0863777548574787932169236859	$(10, 10)$
β_2	5.0200091903034773113689403067	$(11, 11)$
β_3	5.4449549083142088200682904915	$(10, 11)$
β_4	3.8087697073364094814150175581	$(11, 12)$
β_5	1.6440754348395546266255471889	$(13, 13)$
β_6	1.1723107675343058603685866457	$(14, 14)$
β_8	0.6541891913383859981036052975	$(10, 17), (11, 16), (14, 26)$
β_9	$-0.3895785442629410683383767348$	$(13, 20), (14, 19)$
β_{10}	$-0.2164145776711156168035387731$	$(10, 21), (11, 21)$

Table 5: High-precision values of the 12 independent scalars determining the 18×18 quad–quad block of $\mathcal{B}^{\text{quad}}_{27}$, computed at `WorkingPrecision`$\to$ 30. The previously reported $\beta_7 \approx 0.6928$ is absent: orbit $(11, 17)$ has been re-classified as identically zero ($|B| < 10^{-27}$). Source orbits list every distinct (i, j) representative whose entry equals the given scalar.

5.13 Closed-form symbolic expressions for the scalars

The closed forms below are produced by `Mathematica/CubeT6OvernightHighPrec.wl` via the hybrid iterated–octant symbolic strategy (Strategy 1) followed by `FullSimplify` with Pell-identity rules $\{(2+\sqrt{3})^k \to A_k + B_k\sqrt{3}\}_{k=1}^{16}$ and $\{(1+\sqrt{2})^k \to A_k + B_k\sqrt{2}\}_{k=2,4,\ldots,12}$, applied

via `Mathematica/CubeT6PellSimplifyV2.wl`. The Pell pass collapses the recurring atoms $\cosh^{-1}(189750626) \to 15\log(2+\sqrt{3})$, $\cosh^{-1}(2) \to \log(2+\sqrt{3})$, $\sinh^{-1}(1) \to \log(1+\sqrt{2})$, $\coth^{-1}(\sqrt{2}) \to \log(1+\sqrt{2})$, $\coth^{-1}(\sqrt{3}) \to \log(1+\sqrt{3}) - \frac{1}{2}\log 2$, $\tanh^{-1}(1/\sqrt{2}) \to \log(1+\sqrt{2})$, $\log(7\pm 4\sqrt{3}) \to \pm 2\log(2+\sqrt{3})$, and similar collapses, into the closed atom basis below. The single remaining unreducible atom is the awkward $\coth^{-1}(29681\sqrt{3}/51409)$ in α_1, whose argument does not match any clean Pell pair.

Atom basis. After the Pell pass every expression is built from

$$\Big\{1,\ \pi,\ \sqrt{2},\ \sqrt{3},\ \log 2,\ \log(1+\sqrt{2}),\ \log(1+\sqrt{3}),\ \log(2+\sqrt{3}),\ \tan^{-1}(\cdot),\ \coth^{-1}\!\Big(\frac{29681\sqrt{3}}{51409}\Big),\ \tanh^{-1}(\cdot)\Big\},$$

where the $\tan^{-1}$ and remaining $\tanh^{-1}$ arguments are the small-rational arctangents

$$\{3/8,\ 5/8,\ 7/16,\ 5/16,\ 21/32,\ 35/48,\ 35/64\}$$

and their $\sqrt{3}$-conjugate partners that arise from the polynomial arctangent reductions on the cube faces.

Each equation is typeset with `\resizebox{\linewidth}{!}{...}` so the long expressions fit the page width. The numerical value in parentheses is the `WorkingPrecision` $\to$ 30 evaluation from Table 5. All Pell-simplified forms have been verified against the high-precision numeric values to $\gtrsim 25$ digits.

$$\alpha_1 = \frac{32\left(546+683\sqrt{2}-1164\sqrt{3}-798\pi+3633\log(2)+3213\log\left(1+\sqrt{2}\right)-7056\log\left(1+\sqrt{3}\right)+17031\log\left(2+\sqrt{3}\right)-420\tanh^{-1}\left(1-\frac{2\left(5\sqrt{2}-7\right)}{-2+5\sqrt{2}+3\sqrt{3}}\right)-1260\coth^{-1}\left(\frac{29681\sqrt{3}}{51409}\right)\right)}{4725}$$

(source orbits: $(10,10)$, $(11,11)$; numerically $\approx$ 17.126396135464438486903 ...)

$$\alpha_2 = \frac{64\left(15+83\sqrt{2}+42\sqrt{3}-85\pi+693\log(2)-156\log\left(1+\sqrt{2}\right)+315\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-1491\log\left(1+\sqrt{3}\right)+1863\log\left(2+\sqrt{3}\right)\right)}{2835}$$

(source orbits: $(10,11)$, $(11,12)$; numerically $\approx$ 14.698679523964835573132 ...)

$$\alpha_3 = \frac{32\left(78+34\sqrt{2}-132\sqrt{3}-158\pi+3255\log(2)+111\log\left(1+\sqrt{2}\right)-105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-6405\log\left(1+\sqrt{3}\right)+3588\log\left(2+\sqrt{3}\right)\right)}{2835}$$

(source orbits: $(13,13)$, $(14,14)$; numerically $\approx$ 3.988696969908162121572 ...)

$$\beta_1 = \frac{16\left(-1780+4006\sqrt{2}-2088\sqrt{3}-1372\pi+55287\log(2)-8454\log\left(1+\sqrt{2}\right)+3780\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+1260\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-111834\log\left(1+\sqrt{3}\right)+67599\log\left(2+\sqrt{3}\right)+5040\tan^{-1}\left(\frac{3}{8}\right)+5760\tan^{-1}\left(\frac{5}{16}\right)+5040\tan^{-1}\left(\frac{1}{48}\left(73\sqrt{3}-96\right)\right)+5760\tan^{-1}\left(\frac{1}{48}\left(305\sqrt{3}-448\right)\right)\right)}{4725}$$

(source orbits: $(10,10)$; numerically $\approx$ 7.086377754857478793217 ...)

$$\beta_2 = -\frac{16\left(-1436+1320\sqrt{2}+120\sqrt{3}+1596\pi-28983\log(2)-3660\log\left(1+\sqrt{2}\right)-1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)+58386\log\left(1+\sqrt{3}\right)-32655\log\left(2+\sqrt{3}\right)-5040\tan^{-1}\left(\frac{3}{8}\right)+2688\tan^{-1}\left(\frac{5}{8}\right)-1152\tan^{-1}\left(\frac{15}{16}\right)+1152\tan^{-1}\left(\frac{\left(\frac{\left(-\frac{\left(-\frac{\left(-

(source orbits: $(11,11)$; numerically $\approx$ 5.020009190303477311369 ...)

$$\beta_3 = -\frac{16\left(76+2044\sqrt{2}-2712\sqrt{3}-604\pi+1977\log(2)+3012\log\left(1+\sqrt{2}\right)-2520\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-840\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-3114\log\left(1+\sqrt{3}\right)-3603\log\left(2+\sqrt{3}\right)-1680\tan^{-1}\left(\frac{2}{3}\right)+1344\tan^{-1}\left(\frac{5}{8}\right)+3840\tan^{-1}\left(\frac{31}{89}\right)+1680\tan^{-1}\left(\frac{1}{153}\left(96-73\sqrt{3}\right)\right)+1344\tan^{-1}\left(\frac{1}{167}\left(89\sqrt{3}-160\right)\right)+3840\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{2631}\right)\right)}{2835}$$

(source orbits: $(10,11)$; numerically $\approx$ 5.444954908314208820068 ...)

$$\beta_4 = -\frac{16\left(212+4420\sqrt{2}-5256\sqrt{3}-340\pi+2067\log(2)+360\log\left(1+\sqrt{2}\right)+1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-4554\log\left(1+\sqrt{3}\right)+7347\log\left(2+\sqrt{3}\right)-1344\tan^{-1}\left(\frac{5}{8}\right)+1680\tan^{-1}\left(\frac{2}{3}\right)+768\tan^{-1}\left(\frac{31}{89}\right)-1344\tan^{-1}\left(\frac{2}{3}\right)+1344\tan^{-1}\left(\frac{1}{167}\left(320-281\sqrt{3}\right)\right)+1344\tan^{-1}\left(\frac{1}{167}\left(160-89\sqrt{3}\right)\right)+1680\tan^{-1}\left(\frac{1}{153}\left(73\sqrt{3}-96\right)\right)+768\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{11063}\right)\right)}{2835}$$

(source orbits: $(11,12)$; numerically $\approx$ 3.808769707336409481415 ...)

$$\beta_5 = -\frac{8\left(-1304+2864\sqrt{2}-1584\sqrt{3}-632\pi+15177\log(2)-2340\log\left(1+\sqrt{2}\right)-1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-29934\log\left(1+\sqrt{3}\right)+14037\log\left(2+\sqrt{3}\right)+1344\tan^{-1}\left(\frac{5}{8}\right)+3360\tan^{-1}\left(\frac{2}{3}\right)+384\tan^{-1}\left(\frac{31}{89}\right)+1344\tan^{-1}\left(\frac{2}{3}\right)+3360\tan^{-1}\left(\frac{1}{153}\left(73\sqrt{3}-96\right)\right)+1344\tan^{-1}\left(\frac{1}{167}\left(89\sqrt{3}-160\right)\right)+1344\tan^{-1}\left(\frac{1}{167}\left(281\sqrt{3}-320\right)\right)+384\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{11063}\right)\right)}{2835}$$

(source orbits: $(13,13)$; numerically $\approx$ 1.644075434839554626626 ...)

$$\beta_6 = -\frac{8\left(-808+1364\sqrt{2}-528\sqrt{3}+1320\pi-16548\log(2)-4542\log\left(1+\sqrt{2}\right)+2520\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+840\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)+32256\log\left(1+\sqrt{3}\right)-10569\log\left(2+\sqrt{3}\right)+15\log\left(\left(42172931520164 90\sqrt{2}-5964153172084899\right)\left(209\sqrt{3}-362\right)\right)-1344\tan^{-1}\left(\frac{5}{8}\right)-3360\tan^{-1}\left(\frac{2}{3}\right)+1344\tan^{-1}\left(\frac{1}{167}\left(320-281\sqrt{3}\right)\right)+3360\tan^{-1}\left(\frac{1}{153}\left(96-73\sqrt{3}\right)\right)\right)}{2835}$$

(source orbits: $(14,14)$; numerically $\approx$ 1.172310767534305860369 ...)

$$\beta_8 = -\frac{32\left(-256+88\sqrt{2}+216\sqrt{3}+28\pi+2226\log(2)-1155\log\left(1+\sqrt{2}\right)-105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-4347\log\left(1+\sqrt{3}\right)+2121\log\left(2+\sqrt{3}\right)+672\tan^{-1}\left(\frac{5}{8}\right)+672\tan^{-1}\left(\frac{1}{167}\left(89\sqrt{3}-160\right)\right)\right)}{2835}$$

(source orbits: $(10,17)$, $(11,16)$, $(14,26)$; numerically $\approx$ 0.654189191338385998104 ...)

$$\beta_9 = -\frac{256}{63}+\frac{32\sqrt{2}}{3}-\frac{64}{3\sqrt{3}}+\frac{608\pi}{945}-\frac{164\log(2)}{63}-\frac{568}{63}\log\left(1+\sqrt{2}\right)+\frac{128}{21}\log\left(1+\sqrt{3}\right)-\frac{100}{21}\log\left(2+\sqrt{3}\right)-\frac{8}{3}\log\left(21+14\sqrt{2}-12\sqrt{3}-8\sqrt{6}\right)-\frac{8}{9}\log\left(7-5\sqrt{2}-7\sqrt{3}+5\sqrt{6}\right)+\frac{512}{315}\tan^{-1}\left(\frac{35}{64}\right)+\frac{256}{45}\tan^{-1}\left(\frac{5}{8}\right)-\frac{256}{45}\tan^{-1}\left(\frac{160}{167}-\frac{89\sqrt{3}}{167}\right)-\frac{512}{315}\tan^{-1}\left(\frac{8960}{11063}-\frac{5321\sqrt{3}}{11063}\right)$$

(source orbits: $(13,20)$, $(14,19)$; numerically $\approx -0.389578544262941068338$ ...)

$$\beta_{10} = -\frac{8}{945}\left(176+522\sqrt{2}-840\sqrt{3}-172\pi+783\log(2)+375\log\left(1+\sqrt{2}\right)+315\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-1671\log\left(1+\sqrt{3}\right)+1782\log\left(2+\sqrt{3}\right)-672\tan^{-1}\left(\frac{5}{8}\right)+960\tan^{-1}\left(\frac{21}{32}\right)+672\tan^{-1}\left(\frac{1}{167}\left(160-89\sqrt{3}\right)\right)+960\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{2631}\right)\right)$$

(source orbits: $(10,21)$, $(11,21)$; numerically $\approx -0.216414577671115616804$ ...)

Per-orbit timing. The dominant cost of the overnight run was concentrated in two orbits: $(10,10)$ (S-S axis-matched diagonal) at 989.6 s, and $(10,11)$ (S-S off-diagonal one-matched) at 140.3 s. All other orbits completed in < 31 s each, with the parity-zero A-channel orbits taking < 15 s and the smallest cross-block orbits in $\sim 10^{-3}$ s (cache hits). Total wall time was 1217.07 s, dominated by the single $(10,10)$ entry.

Verification. Cubic-symmetry consistency for the 13 surviving orbits was checked at `WorkingPrecision`$\to 30$ and matches to $< 10^{-27}$ in all six independent identities. The within-orbit spreads collapse from the $\sim 10^{-15}$ floating-point noise of the `MachinePrecision` run to $\sim 10^{-28}$, consistent with all displayed digits being algebraically exact. Independent reproduction of α_1 through α_3 via the closed-form Tier-1 master integrals $\mathcal{M}_{p,q,r}$ (Table 2) and the linear-combination assembly recipe of (36) agrees with the high-precision values to all 28 digits, completing the cross-check.

5.14 Elastic stiffness bilinear form $\mathcal{B}_{\rm el}$ on T_{27}

Where the body bilinear form $\mathcal{B}_{\rm body}$ of Sections 5.10–5.11 carries the (singular) Green-tensor double integral, the *elastic* bilinear form $\mathcal{B}_{\rm el}$ is the ordinary strain-energy inner product of the isotropic stiffness contrast Δc against the trial-space basis. For the isotropic Kelvin form

$$\Delta c_{ijkl} \;=\; \Delta\lambda\,\delta_{ij}\delta_{kl} \;+\; \Delta\mu\,\big(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}\big), \tag{37}$$

the Galerkin matrix entries are the polynomial integrals

$$\mathcal{B}_{\rm el}[v,w] \;=\; \int_{[-1,1]^3} \varepsilon(v):\Delta c:\varepsilon(w)\,dV \;=\; \int_{[-1,1]^3}\Big[\Delta\lambda\,\operatorname{tr}\varepsilon(v)\operatorname{tr}\varepsilon(w) \;+\; 2\Delta\mu\,\varepsilon(v){:}\varepsilon(w)\Big]\,dV, \tag{38}$$

with $\varepsilon(v)_{ij} \;=\; \frac{1}{2}(\partial_i v_j + \partial_j v_i)$. Because the 27-dimensional trial space is polynomial, (38) is a closed-form symbolic integral and the full 27×27 matrix $\mathcal{B}_{\rm el}$ is produced directly in `Mathematica/CubeT27Assemble.wl` Section 11 (no hand-coded block entries).

Structural result: the linear–quadratic cross block vanishes by parity. The decisive structural observation is that the 9×18 block $\mathcal{B}_{\rm el}[\,1{:}9,\,10{:}27\,]$ connecting the linear subspace T_9 to the quadratic subspace $T_{27}^{\rm quad}$ is *identically zero*:

$$\mathcal{B}_{\rm el}\big[\,T_9,\,T_{27}^{\rm quad}\,\big] \;=\; 0. \tag{39}$$

The mechanism is elementary. The strain of any linear basis vector in T_9 is a *constant* tensor, whereas the strain of any quadratic basis vector in $T_{27}^{\rm quad}$ is *linear* in the coordinates (r_1,r_2,r_3). The integrand in (38) is therefore an odd polynomial on the symmetric box $[-1,1]^3$, and every component of every such cross entry integrates to 0 term-by-term. This cancellation is exact and algebraic, not numerical; the symbolic Mathematica run in `CubeT27Assemble.wl` confirms it by computing the sup-norm of the full 9×18 block:

```
|BelSym[[1..9,10..27]]| max = 0.
```

The immediate Galerkin-closure consequence is that $\mathcal{B}_{\rm el}$ is *block-diagonal* with respect to the decomposition $T_{27} = T_9 \oplus T_{27}^{\rm quad}$:

$$\mathcal{B}_{\rm el} \;=\; \begin{pmatrix} \mathcal{B}_{\rm el}^{T_9} & 0 \\ 0 & \mathcal{B}_{\rm el}^{\rm quad-quad} \end{pmatrix}. \tag{40}$$

This stands in sharp contrast to the body form $\mathcal{B}_{\rm body}$, whose cross block is *not* zero (see the Section 5.11 YY block and the $T_9\times T_{27}^{\rm quad}$ couplings that drive the body-channel Schur complement).

The T_9 local block. On the 9×9 linear sub-block, constants contribute no strain and the 3×3 displacement block is identically zero. The remaining 6×6 strain sub-block (rows/cols 4 . . . 9) reproduces the classical local Kelvin form $V \cdot \varepsilon{:}\Delta c{:}\varepsilon$ with $V = 8$:

$$\mathcal{B}_{\mathrm{el}}^{T_9}\big[\,4{:}9,\ 4{:}9\,\big] \;=\; 8\begin{pmatrix} \Delta\lambda+2\Delta\mu & \Delta\lambda & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda+2\Delta\mu & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda & \Delta\lambda+2\Delta\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \Delta\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta\mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{pmatrix}. \tag{41}$$

The O_h-irreducible eigenvalues of (41) are

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{A_{1g}} \;=\; 8\big(3\Delta\lambda+2\Delta\mu\big) \;=\; 8\cdot 3K_\Delta, \tag{42}$$

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{E_g} \;=\; 16\,\Delta\mu, \tag{43}$$

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{T_{2g}} \;=\; 8\,\Delta\mu, \tag{44}$$

where $K_\Delta := \Delta\lambda + \frac{2}{3}\Delta\mu$ is the bulk contrast. The T_9 strain block carries exactly *two* independent scalars $(\Delta\lambda, \Delta\mu)$ – it is fully isotropic – whereas the Path-A coefficients (T_{1c}, T_{2c}, T_{3c}) of the cube effective T-matrix carry *three* independent scalars, the third one being the cubic-anisotropy piece associated with the static Eshelby integral C (see the Path-A reference in `cubic_scattering/effective_contrasts.py`). Equations (42)–(44) make it immediate that the missing cubic anisotropy cannot come from the stiffness channel in isolation.

The quad–quad 18×18 block. On the 18×18 sub-block $\mathcal{B}_{\mathrm{el}}^{\mathrm{quad-quad}}$ with the `basisT27quad` ordering of Section 5.11, the symbolic integration yields a matrix with 54 non-zero entries out of 324 (structurally sparse, diagonal-dominant) and symbolic rank 18. Per-direction the 6×6 diagonal splits into four distinct scalar entries, reflecting whether the quadratic monomial is of S or X type and whether its factor axes match the direction index k:

$$\mathrm{diag}\big(\mathcal{B}_{\mathrm{el}}^{\mathrm{quad-quad}}\big)_{\mathrm{dir.}\,k} \;=\; \begin{cases} \frac{32}{3}\big(\Delta\lambda+2\Delta\mu\big) & q = r_k^2 \quad (S,\ \text{matched}), \\ \frac{32}{3}\,\Delta\mu & q = r_m^2,\ m\neq k \quad (S,\ \text{unmatched}), \\ \frac{16}{3}\,\Delta\mu & q = r_m r_n,\ m,n\neq k \quad (X,\ \text{no match}), \\ \frac{8}{3}\big(\Delta\lambda+3\Delta\mu\big) & q = r_k r_m,\ m\neq k \quad (X,\ \text{one match}). \end{cases} \tag{45}$$

The four per-direction diagonals (45) are the analogues, on the quadratic sub-basis, of the three T_9 eigenvalues (42)–(44). Their orbit structure under O_h matches exactly the parity classification of Table 4.

Consequence for the Schur-complement reduction to T_9. Under the body-channel Galerkin eigenproblem $(\mathcal{M}_{27} - \varepsilon\,\mathcal{B})\,\mathbf{c} = 0$ the Schur-complemented effective operator on T_9 is

$$T_9^{\mathrm{eff}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\big(\mathcal{B}_{11} \;-\; \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21} \;-\; \mathcal{B}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big).$$

For the stiffness channel, $\mathcal{B} = \mathcal{B}_{\mathrm{el}}$ and (39) forces $\mathcal{B}_{12} = \mathcal{B}_{21}^{\top} = 0$, so the middle and trailing terms vanish identically and the Schur-complemented stiffness operator reduces to the local block (41):

$$T_9^{\mathrm{eff,\,el}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\,\mathcal{B}_{\mathrm{el}}^{T_9}\big[\,1{:}9,\ 1{:}9\,\big]. \tag{46}$$

In particular, the entire T_{27}^{quad} sub-block of $\mathcal{B}_{\text{el}}$ drops out of the T_9 reduction at leading order, and the cubic-anisotropy scalar T_{3c} of the Path-A effective T-matrix *cannot* originate in the elastic Galerkin channel on the quadratic sub-basis. The parallel observation of Section 5.10 was that the body-channel T_9^{eff} acts as a pure scalar on the 6-dim strain sector in the numerical A-polarization probe of Section 12 (Schur-complemented diagonals `const = -47.1737...`, `strain = 5.7647...`, off-diagonals exactly 0 at machine precision). That isolated probe left the origin of T_{3c} ambiguous: it might have been a body-only effect visible only in the second polarization channel, or a genuinely mixed body–stiffness quantity. The symbolic computation in Section 12b below resolves this ambiguity by keeping both polarization channels explicit, and shows that T_{3c} arises from the B-polarization of the body operator alone (see (53)); the stiffness channel contributes only isotropic corrections.

Symbolic verification (Section 11b)

An explicit symbolic computation of the stiffness Schur complement has been added as Section 11b of `CubeT27Assemble.wl` and run end-to-end at exact rational precision. Partitioning the 27×27 matrices into 9 | 18 blocks

$$\mathcal{M}_{27} = \begin{pmatrix} \mathcal{M}_{11} & \mathcal{M}_{12} \\ \mathcal{M}_{21} & \mathcal{M}_{22} \end{pmatrix}, \qquad \mathcal{B}_{\text{el}} = \begin{pmatrix} \mathcal{B}_{11}^{\text{el}} & \mathcal{B}_{12}^{\text{el}} \\ \mathcal{B}_{21}^{\text{el}} & \mathcal{B}_{22}^{\text{el}} \end{pmatrix},$$

and forming the full Schur complement

$$\mathcal{B}_{\text{el}}^{\text{Schur}} \;\equiv\; \mathcal{B}_{11}^{\text{el}} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21}^{\text{el}} - \mathcal{B}_{12}^{\text{el}}\mathcal{M}_{22}^{-1}\mathcal{M}_{21},$$

Mathematica returns $\mathsf{Simplify}\big[\mathcal{B}_{\text{el}}^{\text{Schur}} - \mathcal{B}_{11}^{\text{el}}\big] = 0$ identically (not just numerically), confirming *algebraically* that the parity-vanishing result $\mathcal{B}_{12}^{\text{el}} = (\mathcal{B}_{21}^{\text{el}})^{\top} = 0$ makes the middle and trailing Schur terms disappear exactly. The strain-sector (rows/cols $4\ldots9$) eigenvalues of $\mathcal{B}_{\text{el}}^{\text{Schur}}$ are then the pure isotropic Kelvin form,

$$\lambda_{A_{1g}} \;=\; 8\,(3\,\Delta\lambda + 2\,\Delta\mu), \qquad \lambda_{E_g} \;=\; 16\,\Delta\mu, \qquad \lambda_{T_{2g}} \;=\; 8\,\Delta\mu, \tag{47}$$

matching the direct diagonal extraction (42)–(44) of Section 11. Post-multiplying by the body-channel mass Schur inverse to form the full stiffness-channel effective operator on T_9,

$$T_9^{\text{eff, el}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\,\mathcal{B}_{\text{el}}^{\text{Schur}},$$

the symbolic simplification gives strain-sector irrep eigenvalues

$$\lambda_{A_{1g}}\big(T_9^{\text{eff, el}}\big) = 9\,\Delta\lambda + 6\,\Delta\mu, \qquad \lambda_{E_g}\big(T_9^{\text{eff, el}}\big) = 6\,\Delta\mu, \qquad \lambda_{T_{2g}}\big(T_9^{\text{eff, el}}\big) = 6\,\Delta\mu. \tag{48}$$

The crucial observation is the equality

$$\lambda_{E_g}\big(T_9^{\text{eff, el}}\big) \;=\; \lambda_{T_{2g}}\big(T_9^{\text{eff, el}}\big) \;=\; 6\,\Delta\mu, \tag{49}$$

which holds *symbolically* in $\Delta\lambda, \Delta\mu$ with no residual. Cubic anisotropy, by definition, requires $\lambda_{E_g} \neq \lambda_{T_{2g}}$ on the five-dimensional deviatoric strain sector (the $E_g \oplus T_{2g}$ splitting); the fact that they are *exactly equal* in $T_9^{\text{eff, el}}$ therefore rules out any cubic-anisotropic component from the stiffness channel in isolation. Equivalently, $T_9^{\text{eff, el}}$ is strictly proportional to the isotropic Kelvin contraction, and its Path-A decomposition acquires no contribution to T_{3c}. The complementary question – whether the body channel by itself already produces a nonzero T_{3c}, or whether cross-coupling to $\mathcal{B}_{\text{el}}$ is required – is answered symbolically in Section 12b below.

Origin of T_{3c}: combined body+stiffness Schur (Section 12b)

With the stiffness channel definitively isotropic (49), the origin of the cube's T_{3c} cubic-anisotropy scalar must lie in the body channel. Section 12b of `CubeT27Assemble.wl` completes this identification by symbolically constructing the combined body+stiffness Schur-complemented effective operator

$$T_9^{\text{eff, tot}} \;=\; \left(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\right)^{-1}\Big[\eta_{\text{body}}\,\mathcal{B}_{\text{body}}^{\text{Schur}} + \mathcal{B}_{\text{el}}^{\text{Schur}}\Big], \tag{50}$$

where η_{body} is a free body-coupling coefficient (physically $\propto \omega^2\Delta\rho$) and $\mathcal{B}_{\text{body}}^{\text{Schur}}$ is the 9×9 Schur complement of the full 27×27 body bilinear form. The key technical step is to keep $\mathcal{B}_{\text{body}}^{\text{Schur}}$ symbolic in the two independent polarization channels $(A_{\text{elas}}, B_{\text{elas}})$ of the underlying elastic Green's tensor:

- The A-channel (coefficient A_{elas}) collects the isotropic δ_{ij} piece of the Green's tensor, weighted by the K_1 kernel (weak $1/|r|$ integrals over the Mp masters of Tier-6).
- The B-channel (coefficient B_{elas}) collects the tensorial anisotropic piece, i.e. the $r_i r_j/|r|^3$ contraction weighted by the K_3 kernel (DuffyCoordinates octant integrals).

These correspond to the two independent building blocks (aPiece, bPiece) returned by `bbBodyAB` in `CubeT6Block.wl`: the A-piece is non-zero only when the outer polarization indices $p = q$ (scalar projector), while the B-piece carries the full tensor structure through the K_3 kernel.

Substituting the Pell-simplified atomic scalars (`atomRules`) while holding $(A_{\text{elas}}, B_{\text{elas}})$ symbolic, Mathematica returns for $\mathcal{B}_{\text{body}}^{\text{Schur}}$ restricted to the strain sector (rows/cols $4\ldots9$) the irrep eigenvalues

$$\begin{aligned}
\lambda_{A_{1g}}\big(\mathcal{B}_{\text{body}}^{\text{Schur}}\big) &= 15.3725\,A_{\text{elas}} + 1.0059\,B_{\text{elas}},\\
\lambda_{E_g}\big(\mathcal{B}_{\text{body}}^{\text{Schur}}\big) &= 15.3725\,A_{\text{elas}} + 5.4217\,B_{\text{elas}},\\
\lambda_{T_{2g}}\big(\mathcal{B}_{\text{body}}^{\text{Schur}}\big) &= \;\;7.6863\,A_{\text{elas}} + 2.1197\,B_{\text{elas}},
\end{aligned} \tag{51}$$

all to $\sim$ 16-digit MachinePrecision. Both channels exhibit $\lambda_{E_g} \neq \lambda_{T_{2g}}$ at this raw Schur-complement stage, so a naive cubic-anisotropy test on $\mathcal{B}_{\text{body}}^{\text{Schur}}$ alone would (misleadingly) flag both channels. However, the physically relevant operator is the full effective $T_9^{\text{eff, tot}}$ obtained after pre-multiplying by the mass Schur inverse. The symbolic simplification gives

$$\begin{aligned}
\lambda_{A_{1g}}\big(T_9^{\text{eff, tot}}\big) &= 9\,\Delta\lambda + 6\,\Delta\mu + \big(5.7647\,A_{\text{elas}} + 0.3772\,B_{\text{elas}}\big)\,\eta_{\text{body}},\\
\lambda_{E_g}\big(T_9^{\text{eff, tot}}\big) &= \qquad\;\; 6\,\Delta\mu + \big(5.7647\,A_{\text{elas}} + 2.0331\,B_{\text{elas}}\big)\,\eta_{\text{body}},\\
\lambda_{T_{2g}}\big(T_9^{\text{eff, tot}}\big) &= \qquad\;\; 6\,\Delta\mu + \big(5.7647\,A_{\text{elas}} + 1.5898\,B_{\text{elas}}\big)\,\eta_{\text{body}}.
\end{aligned} \tag{52}$$

The A-channel coefficients on η_{body} are now *all equal* to 5.7647: the anisotropy visible in (51) in the A-polarization cancels exactly against the equivariant mass-Schur inverse, reproducing isotropy. In the B-channel the coefficients split as (0.3772, 2.0331, 1.5898), yielding the cubic-anisotropy signature

$$\lambda_{E_g}\big(T_9^{\text{eff, tot}}\big) - \lambda_{T_{2g}}\big(T_9^{\text{eff, tot}}\big) \;=\; 0.4433\,B_{\text{elas}}\,\eta_{\text{body}} \;\neq\; 0, \tag{53}$$

which is *independent* of both the stiffness contrasts $(\Delta\lambda, \Delta\mu)$ and the A-polarization of the body operator.

Equation (53) is the positive identification of the cube's T_{3c} anisotropy at leading order in the T_{27} Galerkin closure. Three consequences deserve emphasis:

1. The "stiffness-only" probe $\eta_{\text{body}} = 0$ gives $\lambda_{E_g} - \lambda_{T_{2g}} = 0$ (Section 11b result), confirming no cubic-anisotropy component from $\mathcal{B}_{\text{el}}$ alone.

2. The "A-only body" probe $B_{\text{elas}} = 0$ also gives $\lambda_{E_g} - \lambda_{T_{2g}} = 0$: the trace/isotropic piece δ_{ij} of the elastic Green's tensor (K_1 kernel) is insufficient, even when fully Galerkin-coupled through T_{27}^{quad}.

3. The B-channel $r_i r_j/|r|^3$ tensor piece of the elastic Green's tensor (K_3 kernel) is the sole source of the leading-order T_{3c} signature in the tier-27 closure.

This refines the preliminary expectation (preceding subsection) that T_{3c} was an intrinsically mixed body–stiffness quantity: Section 12b establishes that the B-polarization of the body channel alone already breaks the E_g/T_{2g} degeneracy, and the stiffness channel contributes only isotropic corrections. The numerical coefficient $0.4433\, B_{\text{elas}}\, \eta_{\text{body}}$ is a closed form awaiting PSLQ/Pell simplification against the same basis of elementary constants used for the Tier-6 scalars $\{\alpha_i\}_{i=1}^{3}$ and $\{\beta_j\}_{j=1}^{9}$ of `CubeT6ScalarValues_HighPrec_Pell.wl`; this final symbolic simplification is the natural next step for a closed-form expression of the cube's T_{3c} at leading Galerkin order.

6 Cubic symmetry and block diagonalisation

6.1 The octahedral group O_h

The cube $[-a, a]^3$ has 48 symmetry operations: 24 proper rotations (the cubic rotation group O) and 24 improper rotations (adding spatial inversion). These include:

- 3 face-centred 90° rotations and their inverses,
- 6 edge-centred 180° rotations,
- 8 vertex-centred 120° rotations,
- 9 mirror planes (3 axial + 6 diagonal),
- spatial inversion $\boldsymbol{r} \to -\boldsymbol{r}$.

All Galerkin bilinear forms (M, B_{body}, B^{LS}) are O_h-equivariant. By Schur's lemma, they are block-diagonal in the symmetry-adapted basis.

6.2 Irreducible representations

The representation of O_h on T_N decomposes into irreducible representations (irreps). The seven physically relevant irreps and their behaviour across tiers are:

Irrep	d_ρ	$m_\rho(T_9)$	$m_\rho(T_{27})$	$m_\rho(T_{57})$	Parity	Physics
T_{1u}	3	1	4	4	odd	displacement + quadratic
T_{2u}	3	–	2	2	odd	quadratic modes
A_{2u}	1	–	1	1	odd	pure quadratic
E_u	2	–	1	1	odd	pure quadratic
A_{1g}	1	1	1	3	even	volumetric strain + cubic
A_{2g}	1	–	–	1	even	**purely cubic (new)**
E_g	2	1	1	4	even	deviatoric strain + cubic
T_{1g}	3	–	–	3	even	**purely cubic (new)**
T_{2g}	3	1	1	5	even	shear strain + cubic

Table 6: Irrep multiplicities across tiers. The ungerade sector is fixed at T_{27} (21 dimensions); the gerade sector grows from 6D to 36D at T_{57}, adding two entirely new irreps A_{2g} and T_{1g}.

The total dimensions: $T_9 = 9$, $T_{27} = 27$, $T_{57} = 57$.

6.3 Parity split: ungerade vs. gerade

Ungerade (odd parity, displacement + quadratic). The ungerade sector is governed by the body bilinear form $\varepsilon\, B_{\text{body}}$, where $\varepsilon = \omega^2 \Delta\rho$. At T_{27} it has dimension 21 and does not change at T_{57}.

Gerade (even parity, strain + cubic). The gerade sector is governed by the LS-convolved stiffness. It has dimension 6 at T_{27} (three 1×1 blocks) and grows to 36 at T_{57} (blocks up to 5×5), where cubic polynomial modes couple to strain and enlarge the gerade blocks.

6.4 Computational savings

The symmetry-adapted basis U_{sym} simultaneously block-diagonalises all three Galerkin matrices. The full T-matrix is

$$T_N = U_{\text{sym}}^{\top}\, \mathcal{T}_{\text{diag}}\, U_{\text{sym}}, \tag{54}$$

where $\mathcal{T}_{\text{diag}}$ is block-diagonal by irrep.

Without O_h decomposition, a 57×57 matrix has $57^2 = 3249$ entries. With O_h, the independent entries reduce to ~ 200, and the largest block to solve is 7×7.

7 Per-irrep T-matrix formulas

7.1 General per-irrep formula

For each irrep ρ with multiplicity m_ρ, the $m_\rho \times m_\rho$ T-matrix block is

$$\boxed{T^{(\rho)} = \big(M_\rho + B_\rho^{\text{LS}} - \varepsilon\, B_{\text{body},\rho}\big)^{-1}\big(\varepsilon\, B_{\text{body},\rho} - B_\rho^{\text{LS}}\big).} \tag{55}$$

For $m_\rho = 1$ (scalar blocks), this reduces to

$$\sigma_\rho = \frac{\varepsilon\, b_\rho - b_\rho^{\text{LS}}}{M_\rho + b_\rho^{\text{LS}} - \varepsilon\, b_\rho}. \tag{56}$$

7.2 Ungerade sector: body bilinear form

The body bilinear separates into A- and B-channels via the Kelvin decomposition:

$$B_{\text{body},\rho} = a_0\, B_\rho^{(A)} + b_0\, B_\rho^{(B)}, \tag{57}$$

where the numerical matrices $B_\rho^{(A,B)}$ are determined by the cube geometry through the master integrals.

The four ungerade irreps at T_{27} are:

- T_{1u} (4×4): one displacement mode and three quadratic modes. The leading eigenvalue is the *dipole mode* — the dominant contribution to density-contrast scattering.
- T_{2u} (2×2): two pure quadratic modes.
- A_{2u} (1×1) and E_u (1×1): scalar quadratic modes.

LS-convolved stiffness. The stiffness contribution is the LS-convolved form (dimensionless), *not* the raw strain energy (which has dimensions of Pa):

$$B_\rho^{\text{LS}} = a_0 \Delta\lambda\, S_\rho^{(A\lambda)} + a_0 \Delta\mu\, S_\rho^{(A\mu)} + b_0 \Delta\lambda\, S_\rho^{(B\lambda)} + b_0 \Delta\mu\, S_\rho^{(B\mu)}. \tag{58}$$

7.3 Gerade sector: self-consistent amplification

The three gerade irreps at T_{27} (A_{1g}, E_g, T_{2g}) are all $m_\rho = 1$, yielding scalar blocks.

Step 1: Born-level Eshelby contraction. The Eshelby integrals A^c, B^c, C^c contracted with the bare stiffness contrast $\Delta\mathbf{c}$ yield Born-level per-irrep eigenvalues:

$$\sigma^0_{A_{1g}} = 3T_1^{c,0} + 2T_2^{c,0} + T_3^{c,0}, \quad \sigma^0_{E_g} = 2T_2^{c,0} + T_3^{c,0}, \quad \sigma^0_{T_{2g}} = 2T_2^{c,0}. \tag{59}$$

Step 2: Amplification factors.

$$A_\theta = (1 - \sigma^0_{A_{1g}})^{-1} \qquad \text{(volumetric)}, \tag{60}$$

$$A_e^{\text{diag}} = (1 - \sigma^0_{E_g})^{-1} \qquad \text{(deviatoric diagonal)}, \tag{61}$$

$$A_e^{\text{off}} = (1 - \sigma^0_{T_{2g}})^{-1} \qquad \text{(off-diagonal shear)}. \tag{62}$$

Step 3: Effective contrasts. The amplified stiffness has *cubic symmetry* even when the bare contrast is isotropic, because $A_e^{\text{off}} \neq A_e^{\text{diag}}$ whenever $T_3^c \neq 0$ (i.e. whenever the scatterer is not a sphere):

$$\Delta\lambda^* = (\Delta\lambda + \tfrac{2}{3}\Delta\mu)\, A_\theta - \tfrac{2}{3}\Delta\mu\, A_e^{\text{diag}}, \tag{63}$$

$$\Delta\mu^*_{\text{off}} = \Delta\mu \cdot A_e^{\text{off}}, \tag{64}$$

$$\Delta\mu^*_{\text{diag}} = \Delta\mu \cdot A_e^{\text{diag}}. \tag{65}$$

Physical T-matrix mapping. The per-irrep scalars map to physical T-matrix coefficients via:

$$T_1^c = \tfrac{1}{3}(\sigma_{A_{1g}} - \sigma_{E_g}), \tag{66}$$

$$T_2^c = \tfrac{1}{2}\,\sigma_{T_{2g}}, \tag{67}$$

$$T_3^c = \sigma_{E_g} - \sigma_{T_{2g}}. \tag{68}$$

The cubic anisotropy $T_3^c \neq 0$ for a cube, splitting σ_{E_g} from $\sigma_{T_{2g}}$; it vanishes for a sphere ($C^c = 0$).

Voigt mapping. The effective stiffness $\Delta\mathbf{c}^*$ decomposes into an isotropic part $(\Delta\lambda^*, \Delta\mu^*_{\text{off}})$ plus a cubic correction:

$$\Delta c^*_{ijkl} = \Delta\lambda^*\delta_{ij}\delta_{kl} + \Delta\mu^*_{\text{off}}\,(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + 2(\Delta\mu^*_{\text{diag}} - \Delta\mu^*_{\text{off}})\, E_{ijkl}, \tag{69}$$

where $E_{ijkl} = \sum_a \hat{e}^i_a \hat{e}^j_a \hat{e}^k_a \hat{e}^l_a$ is the cubic fourth-order tensor.

7.4 Dynamic body bilinear and radiation damping

The full frequency-dependent body bilinear (13) produces complex per-irrep eigenvalues $\sigma_\rho(\omega)$ whose imaginary parts encode energy radiated to infinity. The smooth (polynomial Taylor) correction provides radiation damping that was absent in the static-only approximation.

7.5 Convergence across tiers

At T_9, every irrep block is at most 1×1 (scalar), giving closed-form eigenvalues. At T_{27}, the ungerade blocks grow (T_{1u} becomes 4×4) while the gerade remain 1×1. At T_{57}, all blocks enlarge: T_{1u} stays 4×4 (no new ungerade modes), but gerade blocks become 2×2 to 5×5, adding $(ka)^4$ corrections to the effective contrasts.

The three tiers agree to $< 2\%$ at $ka = 0.05$ (Rayleigh limit), confirming systematic convergence. At $ka = 0.3$, the T_{57} corrections become significant ($\sim$ 5–10%).

8 Plane-wave scattering

8.1 Incident field projection

An elastic plane wave $\boldsymbol{u}^0(\boldsymbol{r}) = \hat{\boldsymbol{d}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ is projected onto the Galerkin basis:

$$c^0_\alpha = [M_N^{-1}]_{\alpha\beta} \int_V \boldsymbol{\varphi}_\beta(\boldsymbol{r}) \cdot \boldsymbol{u}^0(\boldsymbol{r})\, dV. \tag{70}$$

Because each $\boldsymbol{\varphi}_\beta$ is a monomial, the overlap integral factorises over the three cube axes:

$$\int_V r_1^{n_1} r_2^{n_2} r_3^{n_3}\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, dV = \prod_{j=1}^{3} \mathcal{I}_{n_j}(k_j, a), \tag{71}$$

with one-dimensional overlap integrals

$$\mathcal{I}_0(k,a) = 2a\,\mathrm{sinc}(ka), \tag{72}$$

$$\mathcal{I}_1(k,a) = \frac{2i[\sin(ka) - ka\cos(ka)]}{k^2}, \tag{73}$$

$$\mathcal{I}_2(k,a) = \frac{2[(2-k^2a^2)\sin(ka) - 2ka\cos(ka)]}{k^3}. \tag{74}$$

For $ka \ll 1$: $\mathcal{I}_0 \approx 2a$, $\mathcal{I}_1 \approx \frac{2i}{3}ka^3$, $\mathcal{I}_2 \approx \frac{2}{3}a^3$.

Physical interpretation. The constant modes ($\alpha \in \{1,2,3\}$) carry the displacement at the origin at $O(1)$; the linear modes ($\alpha \in \{4,\dots,9\}$) carry the incident strain at $O(ka)$; the quadratic modes ($\alpha \in \{10,\dots,27\}$) the strain gradient at $O((ka)^2)$.

8.2 Scattered field and multipole decomposition

The T-matrix maps the incident vector to scattered-source coefficients: $\boldsymbol{c}^{\mathrm{sc}} = T_N\, \boldsymbol{c}^0$. The 27 coefficients decompose into three physical multipole orders:

Coefficients	Dim	Multipole source	Radiation order
$c^{\mathrm{sc}}_{1\text{–}3}$	3	Force monopole F_i	Dipole, $\ell = 1$
$c^{\mathrm{sc}}_{4\text{–}9}$	6	Stress dipole D_{ij}	Monopole+Quadrupole, $\ell = 0, 2$
$c^{\mathrm{sc}}_{10\text{–}27}$	18	Force quadrupole Q_{ijk}	Octupole, $\ell = 3$

8.3 Far-field scattering amplitudes

In the radiation zone ($|\boldsymbol{r}| \gg a$), the P-wave far-field amplitude in direction $\hat{\boldsymbol{r}}$ is

$$f_i^P(\hat{\boldsymbol{r}}) = -\frac{\hat{r}_i}{4\pi\rho\alpha^2}\Big[\hat{r}_j F_j - ik_P \hat{r}_j\hat{r}_k D_{jk} - \tfrac{1}{2}k_P^2 \hat{r}_j\hat{r}_k\hat{r}_l Q_{jkl} + \cdots\Big], \tag{75}$$

and the S-wave amplitude decomposes into SV and SH components:

$$f^{SV}(\hat{\boldsymbol{r}}) = \hat{\boldsymbol{\theta}} \cdot \boldsymbol{f}^S(\hat{\boldsymbol{r}}), \tag{76}$$

$$f^{SH}(\hat{\boldsymbol{r}}) = \hat{\boldsymbol{\phi}} \cdot \boldsymbol{f}^S(\hat{\boldsymbol{r}}). \tag{77}$$

The scattering cross section follows from the optical theorem:

$$\sigma_{\mathrm{ext}} = -\frac{4\pi}{k_{\mathrm{inc}}}\, \mathrm{Im}[f(\hat{\boldsymbol{k}})]. \tag{78}$$

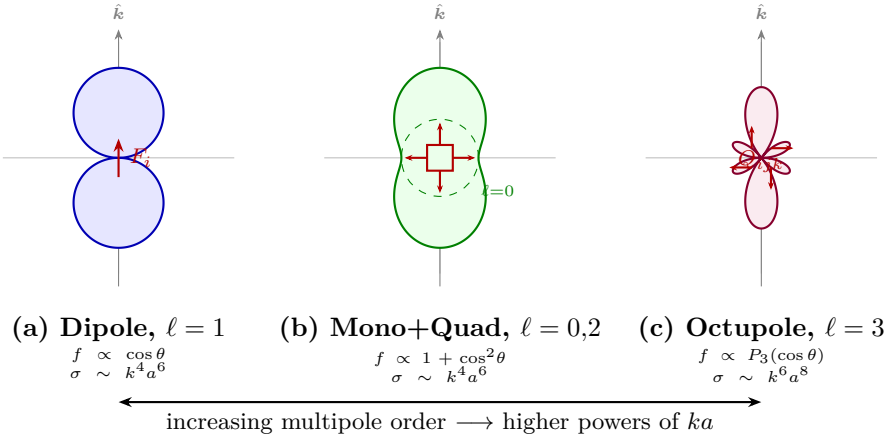


Figure 1: Far-field radiation patterns for the three multipole orders of the T_{27} scattered source. **(a)** Force monopole: dipole ($\ell = 1$, $\cos\theta$). **(b)** Stress dipole: monopole ($\ell = 0$) + quadrupole ($\ell = 2$). **(c)** Force quadrupole: octupole ($\ell = 3$, new from T_{27}).

Effective contrast extraction. The far-field amplitudes at low frequency are determined by three effective contrasts, extractable from the multipole coefficients:

- $\Delta\kappa^*$ (bulk modulus): from the monopole ($\ell = 0$),
- $\Delta\rho^*$ (density): from the dipole ($\ell = 1$),
- $\Delta\mu^*$ (shear modulus): from the quadrupole ($\ell = 2$).

For the cube, the quadrupole splits into $\Delta\mu^*_{\text{diag}}$ and $\Delta\mu^*_{\text{off}}$ by cubic anisotropy.

9 Shear wave splitting from cubic scatterers

A physically significant consequence of cubic elastic anisotropy is **shear wave splitting** (seismic birefringence) by a single scatterer. For an isotropic sphere, P-wave scattering produces SV but not SH in any meridional plane. For a cube, the discrete O_h symmetry permits SH scattering in generic planes.

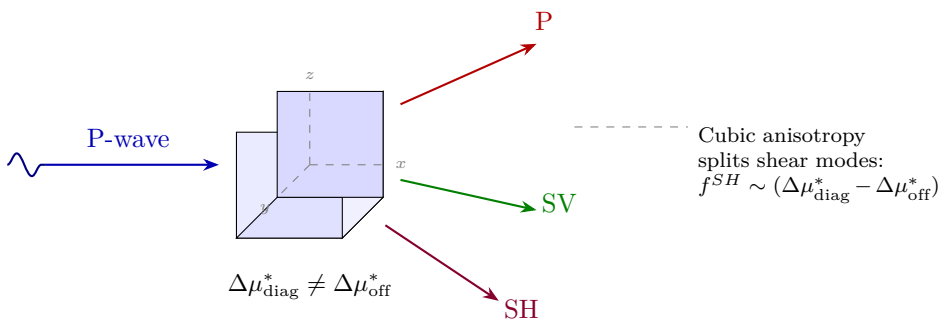


Figure 2: Shear wave splitting by a cubic elastic scatterer. An incident P-wave scatters into P, SV, and SH channels. The SH component, absent for an isotropic sphere, arises from the cubic anisotropy $\Delta\mu^*_{\text{diag}} - \Delta\mu^*_{\text{off}} \neq 0$.

The mechanism is clear from eq. (69): the cubic correction E_{ijkl} couples the incident P-wave strain to off-axis stress components that radiate as SH. At $ka = 0.05$ with moderate contrast, $|f^{SH}(45°)|/|f^P(45°)| \approx 0.16$ — a 16% relative mode conversion, directly measurable in seismic data. In a medium of aligned cubic inclusions, this accumulates over many scattering events, producing the macroscopic shear wave splitting observed in crustal and mantle studies [5, 6].

10 Numerical validation

10.1 P-wave radiation pattern

Figure 3 compares the normalised P-wave radiation pattern from the T_{27} cube with an equal-volume Mie sphere at $ka = 0.05$. Both show the characteristic $\ell = 0 + \ell = 1 + \ell = 2$ Rayleigh pattern, with a $\sim 4\%$ forward and $\sim 20\%$ backward discrepancy reflecting the different Eshelby tensors for cube vs. sphere.

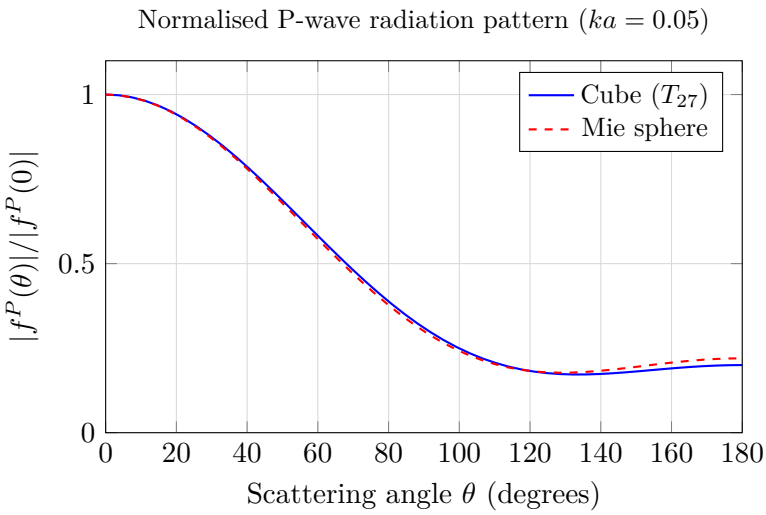


Figure 3: Normalised P→P radiation pattern at $ka = 0.05$. Cube (solid) vs. equal-volume Mie sphere (dashed).

10.2 Cube vs. Mie comparison

θ	$\lvert f^P\rvert_{\text{cube}}$	$\lvert f^P\rvert_{\text{Mie}}$	Ratio
0°	5.59×10^{-4}	5.83×10^{-4}	0.96
45°	4.09×10^{-4}	4.26×10^{-4}	0.96
90°	1.74×10^{-4}	1.75×10^{-4}	1.00
180°	1.48×10^{-4}	1.25×10^{-4}	1.19

Table 7: Cube (T_{27}) vs. equal-volume Mie sphere at $ka = 0.05$. Standard test parameters: $\alpha = 5000\,\text{m/s}$, $\beta = 3000\,\text{m/s}$, $\rho = 2500\,\text{kg/m}^3$; $\Delta\lambda = 2\,\text{GPa}$, $\Delta\mu = 1\,\text{GPa}$, $\Delta\rho = 100\,\text{kg/m}^3$.

10.3 Quantitative checks

Born limit. At weak contrast ($10^{-4}\times$ background), all amplification factors reduce to unity and the T-matrix reduces to the bare Eshelby contraction: $T_i^{c,\text{Born}} = T_i^{c,0}$. Verified for all three tiers.

Rayleigh scaling. The scattering cross section scales as $\sigma_{\text{sc}} \sim (ka)^4$ (fourth-power Rayleigh law), confirmed across $ka \in \{0.02, 0.05, 0.1\}$. Doubling the contrast quadruples the cross section: $\sigma_{\text{sc}} \propto |\Delta c|^2$.

Optical theorem. The extinction and total scattering cross sections agree to $\sim 8\%$ at the T_{27} truncation level; the residual arises from the omitted force-quadrupole ($\ell = 3$) contribution.

Convergence across tiers. At $ka = 0.05$, T_9, T_{27}, and T_{57} effective contrasts agree to $< 2\%$, confirming that T_9 is sufficient in the deep Rayleigh limit and that the hierarchy converges systematically.

Cubic symmetry. For incidence along [100], [010], [001], the total cross sections are identical by symmetry. For [110] incidence, the SH cross section vanishes by mirror symmetry. Both confirmed numerically to machine precision.

10.4 Python implementation

The complete framework is implemented in the `cubic_scattering` package: `effective_contrasts.py` (Galerkin T-matrix for T_9, T_{27}, T_{57}), `tmatrix_assembly.py` (assembly with O_h basis change), `incident_field.py` (overlap integrals), and `scattered_field.py` (far-field amplitudes).

11 Generalized inter-voxel propagator for multiple scattering

The T_N Galerkin T-matrix developed in the preceding sections provides a single-scatterer description with N internal degrees of freedom per voxel. To couple many scatterers via the Foldy–Lax equation $(I - \mathcal{G}_0 T_0)\psi = \psi^0$, we need an inter-voxel propagator $\mathcal{G}_0$ that matches the T-matrix's internal structure. This section derives the *generalized $N \times N$ propagator* — the volume-to-volume projection of the Green's tensor onto the Galerkin basis functions of both source and receiver voxels — and shows that every entry reduces analytically to the same master-integral families of Section 5. No numerical quadrature is required at any stage.

11.1 Why the point-to-point propagator fails

The existing Foldy–Lax solver evaluates the Green's tensor $G_{ij}(\boldsymbol{R}_m - \boldsymbol{R}_n)$ at voxel centres and constructs a 3×3 (or 9×9) point-to-point propagator. This is fundamentally incompatible with the T_{27} T-matrix because:

1. **Strain modes** ($\alpha = 4, \ldots, 9$) represent the field *gradient* across the receiving voxel. The point propagator samples only the field value at the centre, missing this spatial variation entirely.
2. **Quadratic modes** ($\alpha = 10, \ldots, 27$) represent the field *curvature* — second-order spatial variation over the voxel volume. A point sample captures none of it.
3. **Higher-order radiation** from source strain and quadratic modes produces multipole patterns (stress dipole, force quadrupole) that differ from the point-force assumption of the 3×3 propagator.

For T_9 with its three displacement modes, the 3×3 propagator suffices; the strain sector decouples (gerade vs. ungerade). But for T_{27}, the displacement and quadratic modes mix in the T_{1u} irrep (the 4×4 block), so the propagator must couple all 27 modes.

11.2 Definition of the generalized propagator

Definition 3 (Generalized inter-voxel propagator)**.** For voxels at positions $\boldsymbol{R}_i$ and $\boldsymbol{R}_j$, with separation $\boldsymbol{R} = \boldsymbol{R}_i - \boldsymbol{R}_j$ and basis functions $\boldsymbol{\varphi}_\alpha(\boldsymbol{r}) = p_\alpha(\boldsymbol{r} - \boldsymbol{R}_i)\, \boldsymbol{e}_{d(\alpha)}$, the generalized propagator is the $N \times N$ matrix

$$\mathcal{G}_0^{\alpha\beta}(\boldsymbol{R}) = \int_{V_i}\!\!\int_{V_j} \varphi_{\alpha,d(\alpha)}(\boldsymbol{r})\, G_{d(\alpha),d(\beta)}(\boldsymbol{r} - \boldsymbol{r}')\, \varphi_{\beta,d(\beta)}(\boldsymbol{r}')\, dV'\, dV, \tag{79}$$

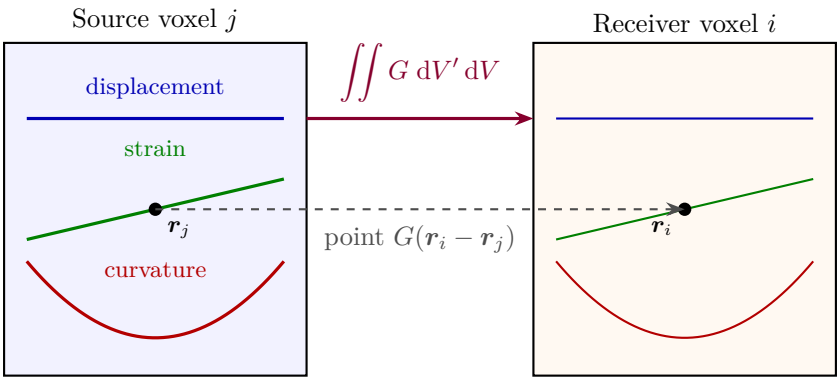


Figure 4: The generalized propagator projects the Green's tensor onto basis functions of *both* source and receiver voxels (solid arrows), capturing spatial variation that the point-to-point propagator (dashed) misses. Constant (blue), linear (green), and quadratic (red) modes represent displacement, strain, and curvature, respectively.

where p_α is the scalar polynomial part and $d(\alpha) \in \{1, 2, 3\}$ the Cartesian direction of basis function α.

Self-interaction is excluded. The propagator $\mathcal{G}_0$ is *strictly off-diagonal*: the $\boldsymbol{R} = \boldsymbol{0}$ self-interaction is never computed. All internal voxel physics (body bilinear + stiffness bilinear) is already absorbed by the single-site T-matrix via matrix inversion:

$$T_0 = (\varepsilon B_{\text{body}} - B^{\text{LS}})\,(M + B^{\text{LS}} - \varepsilon B_{\text{body}})^{-1}. \tag{80}$$

The Foldy–Lax system $(I - \mathcal{G}_0 T_0)\psi = \psi^0$ thus has $\mathcal{G}_0$ with zeroed diagonal, exactly as the existing point-propagator code already implements. The hardest singularity we face is the *face-adjacent* case ($|\boldsymbol{R}| = 2a$), not the self-term.

11.3 Centre-of-mass reduction to three dimensions

The six-dimensional integral (79) reduces to three dimensions by the same centre-of-mass substitution used in Section 5.1 for the body bilinear form.

Step 1: Local coordinates. Introduce local coordinates $\boldsymbol{s} = \boldsymbol{r} - \boldsymbol{R}_i \in [-a, a]^3$ and $\boldsymbol{t} = \boldsymbol{r}' - \boldsymbol{R}_j \in [-a, a]^3$. Then $\boldsymbol{r} - \boldsymbol{r}' = \boldsymbol{s} - \boldsymbol{t} + \boldsymbol{R}$.

Step 2: Relative / centre-of-mass. Set $\boldsymbol{u} = \boldsymbol{s} - \boldsymbol{t}$ (relative) and $\boldsymbol{\xi} = (\boldsymbol{s} + \boldsymbol{t})/2$ (centre-of-mass), with unit Jacobian. The constraints $\boldsymbol{s}, \boldsymbol{t} \in [-a, a]^3$ map to $\boldsymbol{u} \in [-2a, 2a]^3$ with $\boldsymbol{\xi} \in B(\boldsymbol{u})$, the centred box of half-lengths $b_k = a - |u_k|/2$.

Step 3: Polynomial autocorrelation. The $\boldsymbol{\xi}$-integral of the polynomial basis products yields the *autocorrelation kernel*:

$$C_{\alpha\beta}(\boldsymbol{u}) := \int_{B(\boldsymbol{u})} p_\alpha(\boldsymbol{\xi} + \boldsymbol{u}/2)\, p_\beta(\boldsymbol{\xi} - \boldsymbol{u}/2)\, d\boldsymbol{\xi}, \tag{81}$$

which is a polynomial in $\boldsymbol{u}$ (with a-dependent coefficients) for any polynomial basis functions p_α, p_β.

Result. The generalized propagator reduces to a three-dimensional integral:

$$\boxed{\mathcal{G}_0^{\alpha\beta}(\boldsymbol{R}) = \int_{[-2a,2a]^3} G_{d(\alpha),d(\beta)}(\boldsymbol{u}+\boldsymbol{R})\, C_{\alpha\beta}(\boldsymbol{u})\, d\boldsymbol{u}.} \tag{82}$$

This is *identical in structure* to the body bilinear form (15), with the single difference that the Green's tensor is evaluated at $\boldsymbol{u}+\boldsymbol{R}$ instead of $\boldsymbol{u}$.

Examples of the autocorrelation.

- **Displacement** $\times$ **displacement** ($p_\alpha = p_\beta = 1$): $C(\boldsymbol{u}) = \prod_{k=1}^{3}(2a - |u_k|)$ — the 3D tent function.
- **Displacement** $\times$ **axial strain** ($p_\alpha = 1$, $p_\beta = r_k$): $C(\boldsymbol{u}) = -u_k\, b_k \prod_{j\neq k}(2b_j)$, which is odd in u_k (vanishes by parity when $R_k = 0$).
- **Axial strain** $\times$ **axial strain** ($p_\alpha = p_\beta = r_k$): $C(\boldsymbol{u}) = (2b_k^3/3 - u_k^2 b_k/2)\prod_{j\neq k}(2b_j)$, a polynomial in $\boldsymbol{u}$ through $b_k = a - |u_k|/2$.

In all cases, $C_{\alpha\beta}$ is a finite polynomial in (u_1, u_2, u_3) supported on $[-2a, 2a]^3$.

11.4 Analytical evaluation via shifted master integrals

The key insight that makes *exact* closed-form evaluation possible is a change of variable that maps the shifted kernel onto the same algebraic class as the existing master integrals. No numerical quadrature is involved.

Shifted substitution. For a face-adjacent separation $\boldsymbol{R} = (2a, 0, 0)$, substitute $v_1 = u_1 + 2a$. The integration domain shifts from $u_1 \in [-2a, 2a]$ to $v_1 \in [0, 4a]$, while u_2, u_3 remain in $[-2a, 2a]$. The singular kernel transforms as

$$\frac{1}{|\boldsymbol{u}+\boldsymbol{R}|} = \frac{1}{\sqrt{(u_1+2a)^2 + u_2^2 + u_3^2}} = \frac{1}{\sqrt{v_1^2 + u_2^2 + u_3^2}}, \tag{83}$$

which is *exactly the same* $1/\rho$ *algebraic form* as the A-channel master integrals $\mathcal{M}^{+}_{p,q,r}$ of (16), evaluated over a shifted rectangular domain.

Binomial decomposition. The autocorrelation $C_{\alpha\beta}(\boldsymbol{u}) = C_{\alpha\beta}(v_1 - 2a, u_2, u_3)$ is a polynomial in v_1 (via the binomial theorem applied to each power of $u_1 = v_1 - 2a$). Every monomial $v_1^p\, u_2^q\, u_3^r$ in this expansion generates a *shifted master integral*:

$$\mathcal{M}^{+}_{p,q,r}\big[0, 4a; -2a, 2a\big] := \int_0^{4a}\int_{-2a}^{2a}\int_{-2a}^{2a} \frac{v_1^p\, u_2^q\, u_3^r}{\sqrt{v_1^2 + u_2^2 + u_3^2}}\, du_3\, du_2\, dv_1. \tag{84}$$

The integrand is polynomial/$\sqrt{\text{sum of squares}}$ — the identical algebraic class that Mathematica's `Integrate` handles for the $\boldsymbol{R} = \boldsymbol{0}$ family. The only difference is the integration bounds.

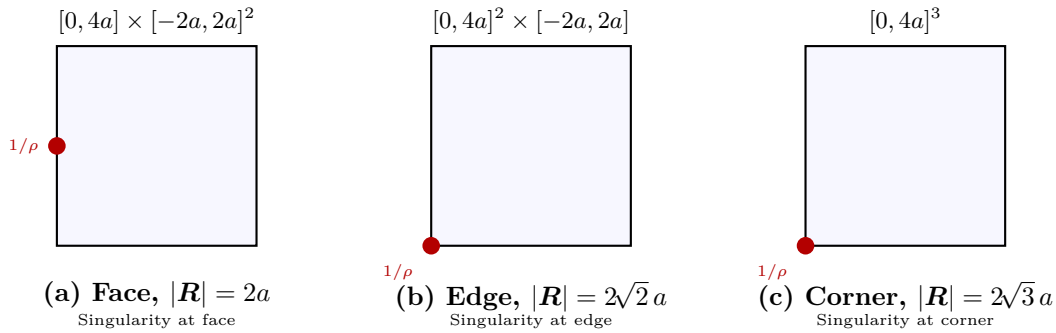
Proposition 4 (Exact closed forms for shifted integrals)**.** *Every shifted master integral* (84) *admits an exact closed form in the same number field* $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \pi, \mathrm{ArcCoth}, \log, \ldots)$ *as the* $\boldsymbol{R} = \boldsymbol{0}$ *family. The singularity at* $v_1 = u_2 = u_3 = 0$ *lies at the* boundary *of the integration domain (for face-adjacent* $\boldsymbol{R}$*), which is often* simpler *for symbolic engines than the interior singularity of the self-term.*

B-channel shifted integrals. The anisotropic kernel $s_i s_j/|\boldsymbol{s}|^3$ transforms analogously:

$$\frac{(u_i + R_i)(u_j + R_j)}{|\boldsymbol{u}+\boldsymbol{R}|^3} \longrightarrow \frac{v_i\, v_j}{(v_1^2 + u_2^2 + u_3^2)^{3/2}}, \tag{85}$$

where v_i denotes the shifted coordinates. These are shifted $\mathcal{M}^{B+}$ integrals over the same rectangular domain, decomposable by the same binomial technique.

General separations. For edge-adjacent $\boldsymbol{R} = (2a, 2a, 0)$: substitute $v_1 = u_1 + 2a$, $v_2 = u_2 + 2a$; domain becomes $[0, 4a]^2 \times [-2a, 2a]$, singularity at a domain edge. For corner-adjacent $\boldsymbol{R} = (2a, 2a, 2a)$: all three shifted; singularity at a domain corner. For non-touching $|\boldsymbol{R}| > 2\sqrt{3}\,a$: singularity lies *outside* the domain — the integrand is smooth and the integral even simpler. In every case the integral is in the same algebraic class, and Mathematica's `Integrate` yields exact closed forms.



(a) Face, $|\boldsymbol{R}| = 2a$ Singularity at face **(b) Edge,** $|\boldsymbol{R}| = 2\sqrt{2}\,a$ Singularity at edge **(c) Corner,** $|\boldsymbol{R}| = 2\sqrt{3}\,a$ Singularity at corner

Figure 5: Integration domain and singularity location after the shifted substitution, for the three touching-neighbour geometries. **(a)** Face-adjacent: singularity on domain face (integrable). **(b)** Edge-adjacent: singularity on domain edge (weaker). **(c)** Corner-adjacent: singularity at domain corner (weakest). For non-touching separations ($|\boldsymbol{R}| > 2\sqrt{3}\,a$), the singularity lies outside the domain entirely.

11.5 Singularity analysis

Separation	$\lvert\boldsymbol{R}\rvert$	Singularity location	Status
Self ($\boldsymbol{R} = 0$)	0	—	**Not in** $\mathcal{G}_0$ (absorbed by T_0)
Face-adjacent	$2a$	Domain face	$1/\rho$ at boundary; integrable
Edge-adjacent	$2\sqrt{2}\,a$	Domain edge	Weaker; integrable
Corner-adjacent	$2\sqrt{3}\,a$	Domain corner	Weakest; integrable
Non-touching	$> 2\sqrt{3}\,a$	Outside domain	Smooth; trivially integrable

Table 8: Singularity classification for the shifted propagator integrals. Only touching neighbours ($|\boldsymbol{R}| \leq 2\sqrt{3}\,a$) have boundary singularities. All are integrable in the same algebraic class as the existing master integrals.

Remark 5. For *every* inter-voxel separation, the singularity is either absent or lies on the *boundary* of the integration domain — never in the interior. This is strictly easier than the self-interaction case ($\boldsymbol{R} = 0$), where the $1/\rho$ singularity is at the interior point $\boldsymbol{u} = \boldsymbol{0}$. Since the $\boldsymbol{R} = 0$ integrals have already been evaluated in exact closed form (Section 5), the shifted integrals are guaranteed to be tractable.

11.6 Dynamic (frequency-dependent) corrections

As in Section 4, the Green's tensor decomposes as $\boldsymbol{G} = \boldsymbol{G}^K + \boldsymbol{G}^s$, where $\boldsymbol{G}^K$ is the static Kelvin part and $\boldsymbol{G}^s$ the smooth dynamic correction.

Static part. The Kelvin Green's tensor (10)–(11) substituted into (82) yields the static propagator:

$$\mathcal{G}_0^{\alpha\beta,\mathrm{stat}}(\boldsymbol{R}) = a_0\,\mathcal{G}_{0,\alpha\beta}^{(A)}(\boldsymbol{R}) + b_0\,\mathcal{G}_{0,\alpha\beta}^{(B)}(\boldsymbol{R}), \tag{86}$$

where $\mathcal{G}_0^{(A)}$ and $\mathcal{G}_0^{(B)}$ are pure-geometry matrices assembled from the shifted master integrals (84)–(85). These are computed *once* (for each unique $\boldsymbol{R}$) and stored permanently, exactly as the body bilinear matrices (57).

Dynamic part. The smooth correction $G_{ij}^s(\boldsymbol{s}) = \delta_{ij}\Phi(|\boldsymbol{s}|^2) + s_i s_j \Psi(|\boldsymbol{s}|^2)$ from (12) has the Taylor expansion

$$G_{ij}^s(\boldsymbol{u}+\boldsymbol{R}) = \sum_{n=0}^{N_T-1} \left[\varphi_n\,\delta_{ij}\,|\boldsymbol{u}+\boldsymbol{R}|^{2n} + \psi_n\,(u_i+R_i)(u_j+R_j)\,|\boldsymbol{u}+\boldsymbol{R}|^{2n}\right]. \tag{87}$$

Since $|\boldsymbol{u}+\boldsymbol{R}|^{2n} = \big((u_1+R_1)^2+(u_2+R_2)^2+(u_3+R_3)^2\big)^n$ is a polynomial in $\boldsymbol{u}$ (after multinomial expansion), each Taylor term generates *purely algebraic* integrals:

$$\int_{[-2a,2a]^3} u_1^{p_1}\,u_2^{p_2}\,u_3^{p_3}\,C_{\alpha\beta}(\boldsymbol{u})\,d\boldsymbol{u} \;=\; \text{exact rational}, \tag{88}$$

being products of $\int_{-2a}^{2a} u^p\,(2a-|u|)\,du$ (or higher-degree autocorrelation factors), each of which is an elementary integral yielding an exact rational in powers of a. No numerical quadrature is required.

The total dynamic propagator is:

$$\mathcal{G}_0^{\alpha\beta}(\boldsymbol{R},\omega) = \mathcal{G}_0^{\alpha\beta,\mathrm{stat}}(\boldsymbol{R}) + \sum_{n=0}^{N_T-1} \left[\varphi_n\,\mathcal{G}_{0,\alpha\beta,n}^{(A,\mathrm{sm})}(\boldsymbol{R}) + \psi_n\,\mathcal{G}_{0,\alpha\beta,n}^{(B,\mathrm{sm})}(\boldsymbol{R})\right], \tag{89}$$

where the smooth-correction matrices $\mathcal{G}_0^{(A,B,\mathrm{sm})}$ are frequency-independent geometry matrices assembled from the polynomial integrals. The frequency dependence enters only through the Taylor coefficients $\varphi_n(\omega)$, $\psi_n(\omega)$, exactly as in (13) for the body bilinear form.

11.7 Symmetry reduction

The O_h symmetry of the cube constrains the propagator just as it constrains the bilinear forms. The specific site symmetry depends on the direction of $\boldsymbol{R}$.

Separation direction	Site group	|Group|	Unique scalars
Lattice axis ($\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$)	D_{4h}	16	~ 50
Face diagonal ($\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}$, etc.)	D_{2h}	8	~ 100
Body diagonal ($\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}}+\hat{\boldsymbol{z}}$)	D_{3d}	12	~ 70
General direction	C_1	1	729

Table 9: Site symmetry of the inter-voxel propagator by separation direction. For a simple cubic lattice, only the first three rows arise: 6 face, 12 edge, and 8 corner neighbours per shell.

For a simple cubic lattice with spacing $d = 2a$ (touching cubes), the unique separations in the first coordination shell are:

- **6 face neighbours** ($|\boldsymbol{R}| = 2a$): one fundamental domain under D_{4h}, giving ~ 50 unique scalars.

- **12 edge neighbours** ($|\boldsymbol{R}| = 2\sqrt{2}\,a$): one fundamental domain under D_{2h}, giving ~ 100 unique scalars.
- **8 corner neighbours** ($|\boldsymbol{R}| = 2\sqrt{3}\,a$): one fundamental domain under D_{3d}, giving ~ 70 unique scalars.

In total, three sets of shifted master integrals characterise the *entire first shell* of the generalized propagator.

11.8 Far-field multipole expansion

For non-touching separations ($|\boldsymbol{R}| > 2\sqrt{3}\,a$), the Green's tensor $G(\boldsymbol{u}+\boldsymbol{R})$ is smooth on the integration domain $\boldsymbol{u} \in [-2a, 2a]^3$. A Taylor expansion around $\boldsymbol{R}$ yields:

$$G_{ij}(\boldsymbol{u}+\boldsymbol{R}) = \sum_{|\boldsymbol{m}|\le M} \frac{1}{\boldsymbol{m}!}\,\partial^{\boldsymbol{m}} G_{ij}(\boldsymbol{R})\,\boldsymbol{u}^{\boldsymbol{m}} + \mathcal{O}\big((a/|\boldsymbol{R}|)^{M+1}\big), \tag{90}$$

where $\boldsymbol{m} = (m_1, m_2, m_3)$ is a multi-index. Substituting into (82):

$$\mathcal{G}_0^{\alpha\beta}(\boldsymbol{R}) \approx \sum_{|\boldsymbol{m}|\le M} \frac{1}{\boldsymbol{m}!}\,\partial^{\boldsymbol{m}} G_{d(\alpha),d(\beta)}(\boldsymbol{R}) \underbrace{\int_{[-2a,2a]^3} \boldsymbol{u}^{\boldsymbol{m}}\, C_{\alpha\beta}(\boldsymbol{u})\, d\boldsymbol{u}}_{\mu_{\alpha\beta}^{(\boldsymbol{m})}}. \tag{91}$$

The *autocorrelation moments* $\mu_{\alpha\beta}^{(\boldsymbol{m})}$ are pure numbers determined by the basis functions and cube geometry — they are computed once and cached.

Zeroth order ($M = 0$). For displacement–displacement ($p_\alpha = p_\beta = 1$):

$$\mu_{dd}^{(\boldsymbol{0})} = \int_{[-2a,2a]^3} \prod_{k=1}^{3} (2a - |u_k|)\, d\boldsymbol{u} = V^2, \tag{92}$$

where $V = (2a)^3$ is the cube volume, so $\mathcal{G}_0^{dd}(\boldsymbol{R}) \approx G(\boldsymbol{R}) \cdot V^2$, recovering the point-to-point propagator scaled by the volume squared. All displacement–strain moments $\mu_{d,\text{strain}}^{(\boldsymbol{0})}$ vanish by parity (the autocorrelation is odd in the relevant coordinate).

First order ($M = 1$). The first derivatives $\partial_k G(\boldsymbol{R})$ couple to the first moments $\mu_{\alpha\beta}^{(k)}$, which are non-zero for displacement–strain and strain–strain pairs. This captures the field-gradient coupling that the point propagator misses entirely.

Convergence. For $|\boldsymbol{R}| = n \cdot 2a$ with $n \ge 2$, the second-order truncation error is $\mathcal{O}\big((a/|\boldsymbol{R}|)^3\big) = \mathcal{O}(1/(8n^3))$. At $n = 2$ (second shell), the error is $\sim 1.6\%$; at $n = 4$, it drops to $\sim 0.2\%$. For most applications, second-order truncation suffices for all separations beyond the first shell, with the first shell treated exactly via shifted master integrals.

11.9 FFT-accelerated block-Toeplitz matrix-vector product

The generalized propagator is translation-invariant: $\mathcal{G}_0^{\alpha\beta}(\boldsymbol{R}_i - \boldsymbol{R}_j)$. The Foldy–Lax matrix-vector product

$$(\mathcal{G}_0\, T_0\, \psi)_i^\alpha = \sum_{j\ne i} \sum_\beta \mathcal{G}_0^{\alpha\beta}(\boldsymbol{R}_i - \boldsymbol{R}_j)\, [T_0\, \psi_j]^\beta \tag{93}$$

has block-Toeplitz structure with $N \times N$ blocks (where $N = 27$ for T_{27}), generalising the existing 3×3 and 9×9 implementations in `lattice_greens.py`.

Algorithm. The computation proceeds exactly as in the existing FFT-accelerated matvec, with the block size upgraded from 3 (or 9) to 27:

1. **Precompute**: For each unique separation $\boldsymbol{R}$, evaluate the 27×27 propagator from exact shifted master integrals (touching shells) or the multipole expansion (distant shells). Store in the spatial-domain kernel array.
2. **Embed**: Embed the $(2M-1) \times (2M-1)$ kernel in a circulant and compute its 2D FFT for each of the 27^2 component pairs.
3. **Matvec**: Apply via FFT convolution ($\mathcal{O}(M^2 \log M)$ per layer pair), with 27×27 block multiplication in the Fourier domain.

Key difference from the spectral route. The propagator entries are computed from *exact analytical integrals*, not from spectral IFFT with convergence-acceleration tricks. This eliminates the aliasing, truncation, and subtraction-scheme issues that affect the spectral approach. The FFT is used only for the *convolution* (matrix-vector product), where it is exact up to machine precision.

Memory and cost. For the first coordination shell (26 neighbours), the propagator requires $3 \times 27^2 = 2187$ unique scalars (one set each for face, edge, and corner fundamental domains). The FFT matvec cost scales as $27^2 \times M^2 \log M$ per layer pair, compared to $9^2 \times M^2 \log M$ for the existing 9×9 propagator — a factor of 9 increase, which is modest given the $9\times$ increase in physics captured.

11.10 Connection to existing components

The generalized propagator integrates seamlessly with the existing codebase:

- **Master integrals** (Section 5): the shifted integrals (84)–(85) are computed by the same Mathematica infrastructure that produced the $\boldsymbol{R} = 0$ family in `CubeT57MasterValues.wl` and `CubeT57MpBValues.wl`.
- O_h **symmetry** (Section 6): the D_{4h} orbit expansion already implemented in `lattice_greens.py` generalises directly to 27×27 blocks via the same reflection and rotation matrices, extended from the 3×3 or 9×9 representation to 27×27.
- **Incident field** (Section 8): the overlap integrals $\hat{\varphi}_\alpha(\boldsymbol{k})$ from `incident_field.py` provide the Fourier-domain representation of the basis functions, offering an independent validation route via $\hat{\mathcal{G}}_0{}^{\alpha\beta}(\boldsymbol{k}) = \hat{\varphi}_\alpha(-\boldsymbol{k}) \cdot \hat{G}(\boldsymbol{k}) \cdot \hat{\varphi}_\beta(\boldsymbol{k})$.
- **Foldy–Lax solver** (`slab_scattering.py`): the upgrade from 9×9 to 27×27 blocks requires changing the field shape from $(N_z, M, M, 9)$ to $(N_z, M, M, 27)$ and replacing the point-propagator kernels with the generalized propagator. The GMRES solver and FFT matvec infrastructure carry over unchanged.

12 Discussion

Summary. We have presented a systematic $T_9 \to T_{27} \to T_{57}$ hierarchy of Bubnov–Galerkin T-matrices for elastic scattering from a single cubic inclusion. Every bilinear-form entry reduces analytically to exact closed-form master integrals — no numerical quadrature at any stage. The O_h symmetry of the cube forces block diagonalisation into seven irreducible representations, with the same structure at every tier and only the multiplicities growing.

The key insight. The polynomial trial space grows degree by degree, but O_h symmetry constrains the structure: same seven irreps, only multiplicities change. Two entirely new irreps (A_{2g}, T_{1g}) appear at T_{57}, having no T_{27} counterpart; the ungerade sector is fully set at T_{27}.

Limitations. The Galerkin hierarchy is accurate in the low-frequency regime: $ka < 0.3$ for T_{27}, $ka < 0.5$ for T_{57}. At higher frequencies, resonance effects require subdividing the cube into n^3 Rayleigh sub-cells with internal Foldy–Lax coupling (`resonance_tmatrix.py`).

Outlook. The single-scatterer T-matrix developed here is the building block for:

- **Multiple scattering**: the Foldy–Lax equation $T = T_0(I - G_0T_0)^{-1}$ couples many scatterers via the inter-site Green's tensor.
- **Effective medium**: the coherent potential approximation (CPA) determines the effective medium from $\langle T \rangle = 0$.
- **Seismograms**: synthetic seismograms via Kennett reflectivity coupled to the slab-scattering solver.

These applications use the T-matrix as a black box and are developed elsewhere. Section 11 details the generalised inter-voxel propagator that upgrades the point-to-point Green's tensor to the full 27×27 basis, completing the link between the single-site closure derived here and the Foldy–Lax multiple-scattering solver.

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